1. DDU PROFILE

Dharmsinh DesaiUniversity

ISO 9001:2008

Nadiad - 387 001, Gujarat, India.

University (formerly known as DDIT), was established in June 1968 by Late Shri Dharmsinh Desai (then Member of Parliament), with a view to provide quality education in Engineering and Technology to the youth of Gujarat and was affiliated to Gujarat University.

Having modesty started with Diploma and Degree courses in Chemical Engineering. It has by now grown as top most Institute of Gujarat offering Graduate and Post Graduate level courses in various disciplines, and ultimately has become a State University in the year 2005. The institute gained an autonomous status in 1999 with approval from UGC and Gujarat Government and become a Deemed University accredited by N.B.A. in the year 2000.This University offers Graduate and Post Graduate level coursesin various disciplines and Master of Computer Applications (MCA) is being run under Faculty of Management and Information Science since 1987.

The University has also received ISO-9001:2008 Certification from ISOQAR and have also acquired NAAC accreditation. D.D.University believes not only in providing quality education, but also provides an overall development of its students in the form of being disciplined, sociality accountable to the society and responsible citizen of our county.

D.D. University is being guided by the vision laid down by its founding Chairman Late Shri Dharmsinh Desai, the present Chairman Dr. Narendra Desai and under a strong and dynamic leadership of Vice-chancellor Dr. H.M.Desai.

Vision:
"Become a multi-disciplined & „learner oriented" university to closely associate with and be responsive to the industry, to create supportive and caring environment for staff and students and to engage in R & D activities in areas of national priority."

Mission:
"Undertake programs and projects for development of human resources, both through formal and non-formal delivery systems, in areas of professional pursuits in all walks of human endeavors, with accent on relevance, value addition, societal needs and futuristic pilot projects."

2. COMPANY PROFILE

Company name: Roima Intelligence India private limited

Company address: Satyamev Elite, 608 - 615, Sardar Patel Ring Rd, South Bopal, Bopal, Ahmedabad, Gujarat 380058

Company Email: neha.vindhani@roimaint.com

Company Number: +91 7069692727

About us: Roima Intelligence is a global business-solutions provider specializing in digital transformation for manufacturing, supply chain, logistics, and product lifecycle management.The company delivers a broad portfolio of software and services including ERP, MES/MOM, WMS, PLM, logistics platforms, machine vision, and more to help organizations streamline operations, integrate systems, and manage supply chains effectively.With more than 500 customers worldwide and over 400 professionals working across several countries, Roima brings deep industry expertise and a track record of successful implementations.Roima aims to be a reliable, long-term partner not just a vendor helping clients modernize their infrastructure, improve quality, and operate with greater efficiency and resilience.

Mission: Roima's mission is to deliver powerful, integrated software that helps businesses improve productivity, quality, and efficiency. They focus on creating measurable, long-lasting results through supply-chain, manufacturing, and logistics solutions. Roima works as a long-term partner, guiding clients through digital transformation and ensuring their systems run smoothly, sustainably, and profitably.

Vision: Roima's vision is to lead global digital transformation by helping businesses build smart factories and intelligent supply chains. They aim to make operations fully connected, automated, and data-driven so companies can work faster, smarter, and more efficiently. Roima wants to expand internationally and continue innovating solutions that strengthen competitiveness and prepare organizations for the future.

3. PROJECT PROFILE

Project Title	HR Management System
Company Name	Roima Intelligence India private limited
Tools	Postman, Visual Studio, Vs Code, SQL Server Management Studio
Duration	19th January, 2026 to 21th March, 2026
Team Size	1
Team Member	Kartik Miteshbhai Patel
Internal Guide	Associate Professor Minal V. Shah
External Guide	Yash Shah

4. TOOLS & TECHNOLOGY

Tools :-

Sr. No	Tools	Purpose
1	Visual Studio	Writing Code - Backend
2	Visual Studio Code	Writing Code – Frontend
3	SQL Server Manager Studio	Database
4	Postman	API Testing

Technology :-

Sr. No	Technology	Purpose
1	Asp.net – C#	Backend Technology
2	React.js – TypeScript	Frontend Technology
3	SQL Server Manager Studio	Database

5. PROJECT PLANNING

5.1 Project Definition :

The HR Management System (HRMS) is an online platform designed to simplify and automate various human resource operations within an organization. It provides a centralized system where employees, managers, and HR administrators can manage travel plans, expenses, employee engagement activities, game bookings, and recruitment processes efficiently. The system eliminates manual paperwork and scattered communication by offering a structured digital solution for managing HR-related activities.

The system allows HR/Admin to create and assign travel plans to employees, verify and approve submitted expenses, manage employee records, configure game slots, and manage job postings and referrals. Employees can view assigned travel plans, submit expense claims with supporting documents, participate in engagement activities, book game slots, and refer candidates for job openings. Managers can monitor team activities and organizational hierarchy.

By integrating all HR operations into a single platform, the system improves transparency, reduces processing time, and enhances collaboration between employees, managers, and HR teams. The HRMS also provides real-time status tracking, automated notifications, secure document management, and role-based access control to ensure efficient and secure operations.

Example:

Let's assume that an employee needs to travel for an official meeting. First, the HR/Admin assigns a travel plan through the system. The employee can view the assigned travel details and upload expense receipts after completing the trip. The employee then submits the expense request along with supporting documents.

The **HR/Admin** reviews the submitted expenses and either approves or rejects them with remarks. At the same time, employees can also participate in engagement activities such as posting achievements, liking posts, or booking game slots. Additionally, employees can view job openings and refer candidates through the system.

After approval, the employee can track the expense status in real time, ensuring transparency and smooth communication between employees and HR.

5.2 Preliminary Investigation :

- Traditional HR processes involve manual paperwork and delayed approvals which reduce efficiency.
- Employees often lack a centralized platform to manage travel plans, expenses, engagement activities, and referrals.
- HR teams face difficulty in managing employee data, approvals, and records through scattered systems.
- There is a need for a centralized digital platform to streamline HR operations and improve transparency.
- Automated notifications help users stay informed about travel assignments, expense approvals, and job referrals.
- Role-based access control ensures secure management of employee information and organizational data.

6. SYSTEM ANALYSIS

6.1 Feasibility Study :

6.1.1 Technical Feasibility :

- The proposed HR Management System (HRMS) is technically feasible using a modern Microsoft-based technology stack such as ASP.NET Core Web API, React.js with TypeScript, and SQL Server.
- The backend will be developed using ASP.NET Core Web API, which provides high performance, RESTful API support, dependency injection, and strong security features.
- The frontend will be implemented using React.js with TypeScript, providing a scalable, strongly typed, and maintainable user interface.
- JWT-based authentication and role-based authorization will be used to secure the system and control access for Employees, Managers, and HR administrators.
- Entity Framework Core with SQL Server will be used for ORM-based database access, ensuring data consistency for modules such as travel plans, expenses, achievements, game slots, and job referrals.
- The system supports file upload and document management for receipts, travel documents, CV uploads, and policy files using multipart APIs with validation.
- Email notification services will notify employees and HR regarding travel assignments, expense submissions, referral updates, and system activities.
- Business rules such as expense submission windows, mandatory proof documents, and fair game slot allocation will be implemented in the service layer.
- The backend will support queue-based scheduling logic for fair allocation of game slots among employees.
- Audit logging and status tracking will maintain records of approvals, expense submissions, and referral activities for transparency and compliance.

6.1.2 Operational Feasibility :

- The system provides an easy-to-use interface for HR/Admin, Managers, and Employees, enabling efficient HR operations.
- HR/Admin users can create and manage travel plans, verify and approve expenses, configure game slots, manage job openings, and monitor referrals.
- Employees can view travel plans, submit expenses with proof documents, participate in achievements and celebrations, request game slots, and refer candidates for job openings.
- Managers can view team members, monitor achievements, and access travel and expense information of their direct reports.
- The system provides automated notifications through email and in-app alerts for travel assignments, expense approvals, referrals, and system updates.
- Automated validation and business rules reduce manual errors and ensure efficient workflow management.

6.1.3 Economical Feasibility :

- The proposed HRMS system is economically feasible as it uses open-source or freely available technologies such as ASP.NET Core, React with TypeScript, and SQL Server Express/Developer edition.
- The system reduces operational costs by automating HR processes such as travel management, expense tracking, approvals, referrals, and game slot scheduling.
- Centralized data management improves efficiency and reduces administrative workload for HR teams.
- The modular architecture ensures low maintenance costs and easier system updates.
- The system can be deployed on low-cost cloud or on-premise environments, minimizing infrastructure expenses.
- The platform supports future scalability without significant additional cost, allowing organizations to add new features or users easily.

6.2 System Requirement Specification :

6.2.1 Introduction :

The Human Resource Management System (HRMS) is an internal enterprise web-based application designed to automate and manage core HR operations such as travel management, expense reimbursement, employee engagement, games scheduling, organizational hierarchy visualization, and job referral workflows.

The system aims to reduce manual HR processes, improve transparency, and enhance employee engagement by providing a centralized digital platform accessible to Employees, Managers, and HR personnel.

I. User & Role Management Module
II. Travel & Expense Management Module
III. Social Achievements & Celebrations Module
IV. Games Scheduling Module
V. Organization Chart Module
VI. Job Listing, Sharing & Referral Module

6.2.2 Purpose :

The primary purpose of the HRMS is to provide a secure, centralized, and efficient platform for managing internal HR operations.

The system aims to:

- Automate travel assignments and expense reimbursements
- Reduce manual document handling
- Improve employee engagement through social features
- Ensure fair allocation of shared organizational resources (games)
- Provide transparency in organizational hierarchy
- Simplify job sharing and referral processes

The system helps HR teams:

- Manage employee travel plans and documents
- Track and verify expenses
- Configure job postings and referral workflows
- View organization structure across departments

The system enables employees to:

- View assigned travels and submit expenses
- Upload and manage travel-related documents
- Participate in internal social activities
- Book game slots fairly
- Refer friends for job openings

6.2.3 Problem Statement :

Organizations relying on manual or partially automated HR processes face challenges such as:

- Delays in travel communication and expense reimbursement
- Poor visibility of organizational hierarchy
- Unfair usage of shared recreational resources
- Inefficient job referral tracking
- Lack of employee engagement platforms

The HRMS solves these problems by digitizing and automating HR workflows with role-based access control and centralized data management.

6.2.4 Scope :

6.2.4.1 User Management Module

- **Registration/Login :** Employees, Managers, and HR users can securely log in to the system using authentication credentials.
- **Profile Management :** Users can view and manage their personal profile information within the system.
- **Role-Based Access :** The system supports different roles such as Employee, Manager, and HR, and each role can access features according to their permissions.

6.2.4.2 Travel Management Module

- **Travel Creation and Assignment :** HR can create travel plans and assign them to one or more employees.
- **View Travel Details :**
  Employees can view their assigned travel plans and related information.

6.2.4.3 Expense Management Module

• **Expense Submission :** Employees can submit expenses related to their assigned travel with proof documents.

• **Expense Verification :** HR can verify, approve, or reject submitted expenses and update their status.

6.2.4.4 Achievements & Engagement Module

• **Social Achievements Feed :** Employees can create achievement posts and view posts shared by others.

• **Likes and Comments :** Users can like and comment on achievement posts and participate in engagement activities.

6.2.4.5 Game Slot Management Module

• **Game Slot Scheduling :** HR can configure game slots, including timing, duration, and capacity.

• **Fair Slot Allocation :** Employees can request game slots, and the system allocates them fairly according to defined rules.

6.2.4.6 Organization Chart Module

• **Hierarchy Visualization :** Employees can view the organizational hierarchy and managerial chain within the system.

• **Employee Relationship View :** The system displays reporting relationships between employees and managers.

6.2.4.7 Job Listing & Referral Module

• **Job Listing and Sharing :** HR can create and manage job openings available in the organization.

• **Employee Referrals :** Employees can refer candidates for job openings and track referral status.

6.2.5 Audience and Users :

I. Employee: Views travels, submits expenses, posts achievements, books game slots, refers jobs

II. Manager: View team member, travel documents, expense (read-only), organizational chart

III. Hr: Manager travels, expenses, documents, jobs, referrals, and organizational chart

6.2.6 Overview :

6.2.6.1 Hardware and software (for development)

Hardware Requirements	
Processor	Intel core i5 or higher
RAM	8 GB
Software Requirements	
Backend	ASP.NET Core
Database	SQL Server
Tools	Visual Studio, Visual Studio code, Postman
Browser	Latest versions of Chrome, Edge, Firefox

6.2.6.2 Hardware and software (for User)

Hardware Requirements	
Processor	Intel core i3 or higher
RAM	4 GB
Software Requirements	
Operating System	Windows, Ios, Android
Browser	All Latest Browser

6.2.7 Acronyms and Abbreviations :

- HRMS – Human Resource Management System
- API – Application Programming Interface
- DB – Database
- JWT – JSON Web Token
- RBAC – Role-Based Access Control

6.2.8 Functional Requirements :

6.2.8.1 User & Role Management module :

a. Common Requirements :

- The system shall authenticate users using secure login credentials.
- The system shall enforce role-based access control.
- The system shall store encrypted passwords.

b. Employee :

- The employee shall be able to view and update profile information.
- The employee shall report to a manager and belong to a department.

c. Manager :

- The manager shall be able to view direct reports.
- The manager shall appear correctly in the organization chart.

d. HR :

- HR shall be able to view all employees and departments.
- HR shall manage travel plans, expenses, jobs, and referrals.

6.2.8.2 Travel & Expense Management Module :

a. Travel Assignment :

- HR shall be able to create travel plans for one or more employees.
- The system shall notify employees via email and in-app notification.
- Each travel shall include dates, location, and expense limit.

b. Travel Document :

- The system shall allow HR and employees to upload documents.
- Each document shall store:
  - Travel ID
  - Employee ID
  - Uploaded by
  - Document type
  - File URL
  - Uploaded date
- Permissions:

- HR: View/upload all documents
- Employee: View/upload own documents
- Manager: View team documents (read-only)

c. Expense Submission :

- Expenses shall always be linked to a travel.
- Employees may submit expenses only:
  - After travel start date
  - Within 10 days of travel end date
- The system shall block late submissions.
- Multiple expenses per travel shall be supported.

d. Proof Upload & Verification :

- Each expense must include at least one proof document.
- HR shall be able to approve or reject expenses.
- Rejection requires mandatory remarks.
- Status values: Draft, Submitted, Approved, Rejected

6.2.8.3 Social Achievement & Celebrations Module

a. Achievement Posts :

- Employee Can Create Achievement posts with:
  - Title
  - Description
  - Tags
  - Visibility
- All User can view posts in reverse chronological order
- Users can like and comment on posts.

b. Moderation :

- Employee can edit/delete their posts
- HR can delete inappropriate posts.
- System shall send a warning email on HR moderation

c. Birthday & Anniversary :

- The system shall auto-generate system posts for:
  - Employee Birthday
  - Work Anniversaries
- Auto-generate system posts shall be visible to all users

6.2.8.4 Game Scheduling Module :

a. Configuration :

- HR/Admin can configure:
  - Operating Hours
  - Slot Durations
  - Max and Min Players per slot
- System shall generate daily slot automatically

b. Booking Rules :

- Employee can request slots for game
- One active booking per employee per day
- Fair scheduling shall be enforced.
- Cancellation shall assign next eligible player.

6.2.8.5 Organization Chart Module :

- The system shall display the managerial chain of an employee
- The system shall show one level of direct reports
- Each node shall display:
  - Name
  - Designation
  - Profile image
- Clicking a node reloads the organization chart for that employee

6.2.8.6 Job Listing, Sharing & Referral Module :

a. Job Listing :

- The system shall list active job openings
- Employee can browse job details

b. Job Sharing :

- Employee can share job via email
- The system shall attach the job description document
- Sharing activity shall be logged

c. Job Referrals :

- Employee can refer friends with:
  - Name
  - Email
  - CV

- Note
- The system shall email:
  - HR owner
  - Configured reviewers
- Referral status shall be tracked

6.2.9 Non-Functional Requirements :

6.2.9.1 Accessibility

- The system shall be accessible via modern web browsers.
- UI shall be responsive across devices.

6.2.9.2 Availability

- The system shall be available 24×7 except maintenance.
- Concurrent users shall be supported.

6.2.9.3 Performance

- Frequently accessed data shall be optimized.
- API responses shall be efficient.

6.2.9.4 Security

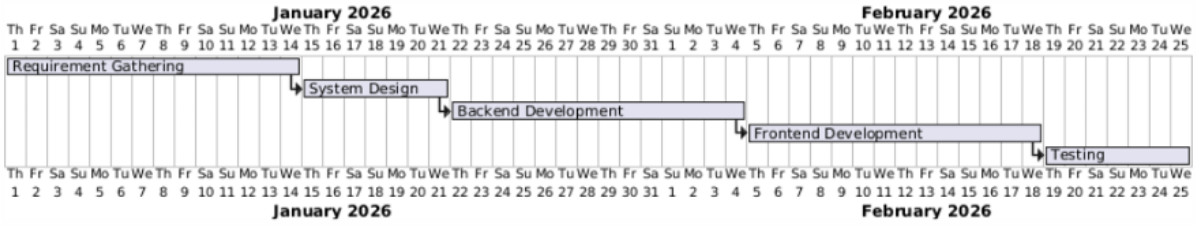
- JWT-based authentication shall be enforced.
- Role-based authorization shall be applied.
- File uploads shall be validated.

6.2.9.5 Maintainability

- Modular architecture shall be followed.
- Database design shall support future expansion.

6.2.10 Other Information :

a. Gantt Chart :

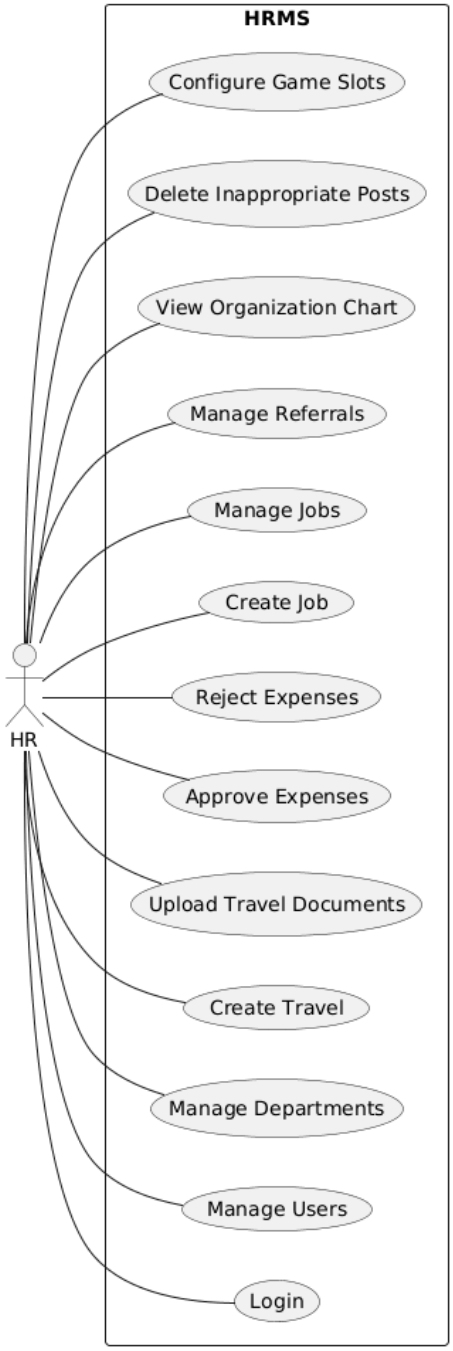


7. SYSTEM DESIGN

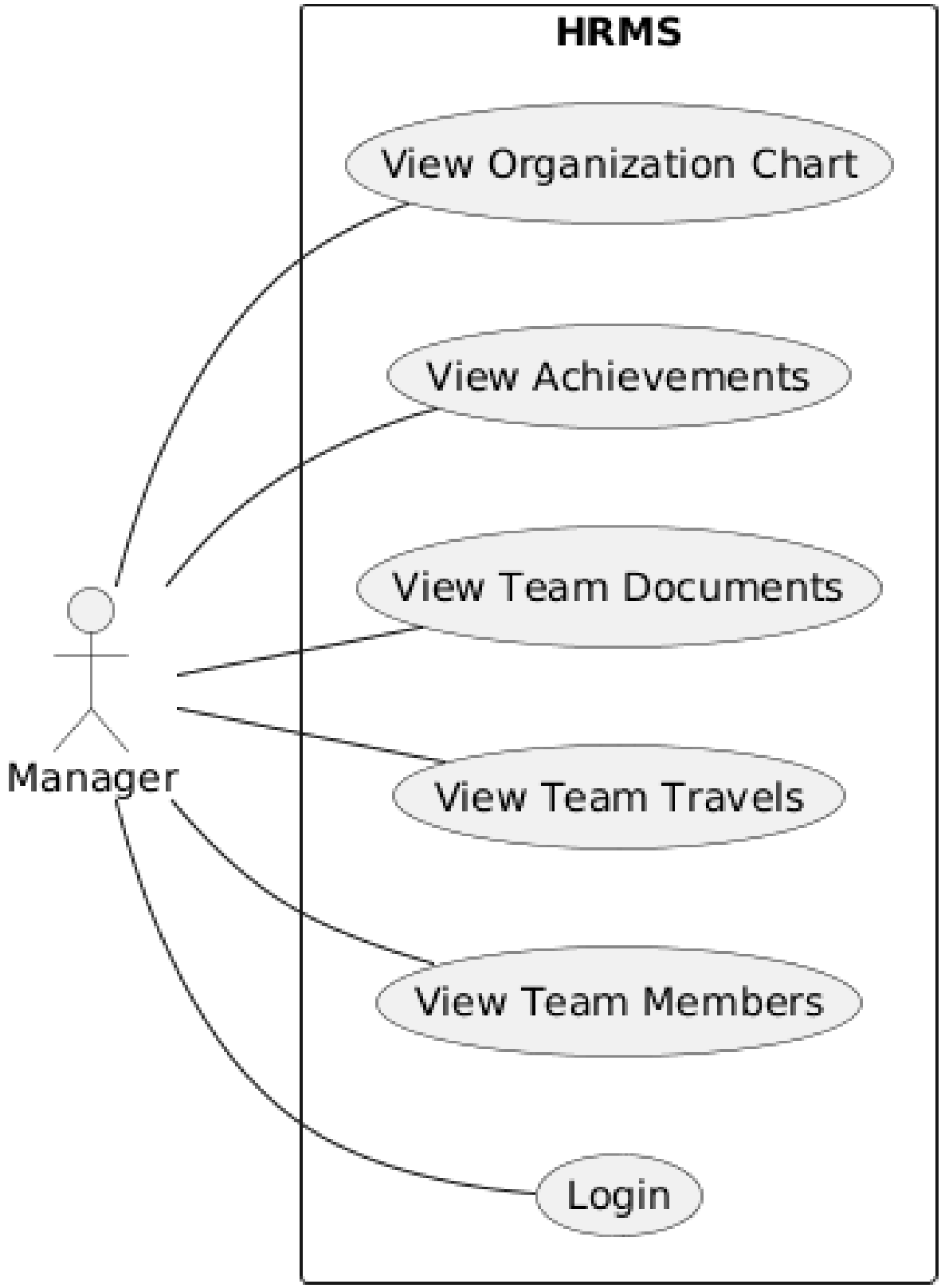
7.1 Architectural Design :

7.1.1 Usecase Diagrams :

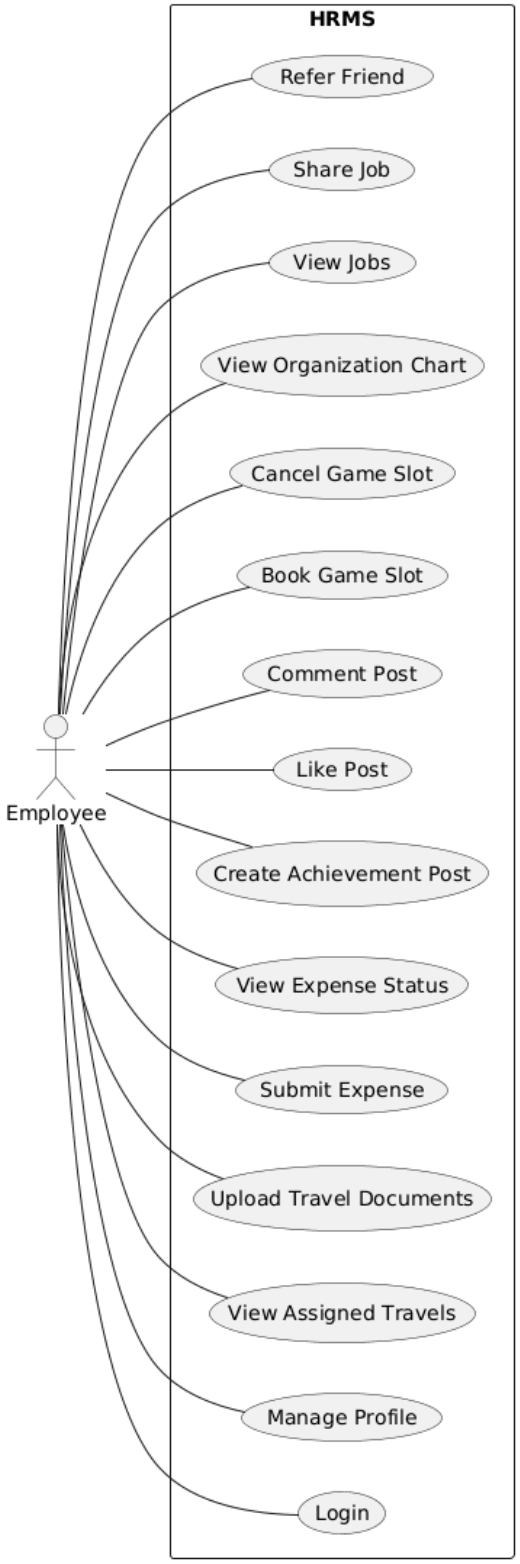
7.1.1.1 HR usecase Diagram:



7.1.1.2 Manager usecase Diagram:

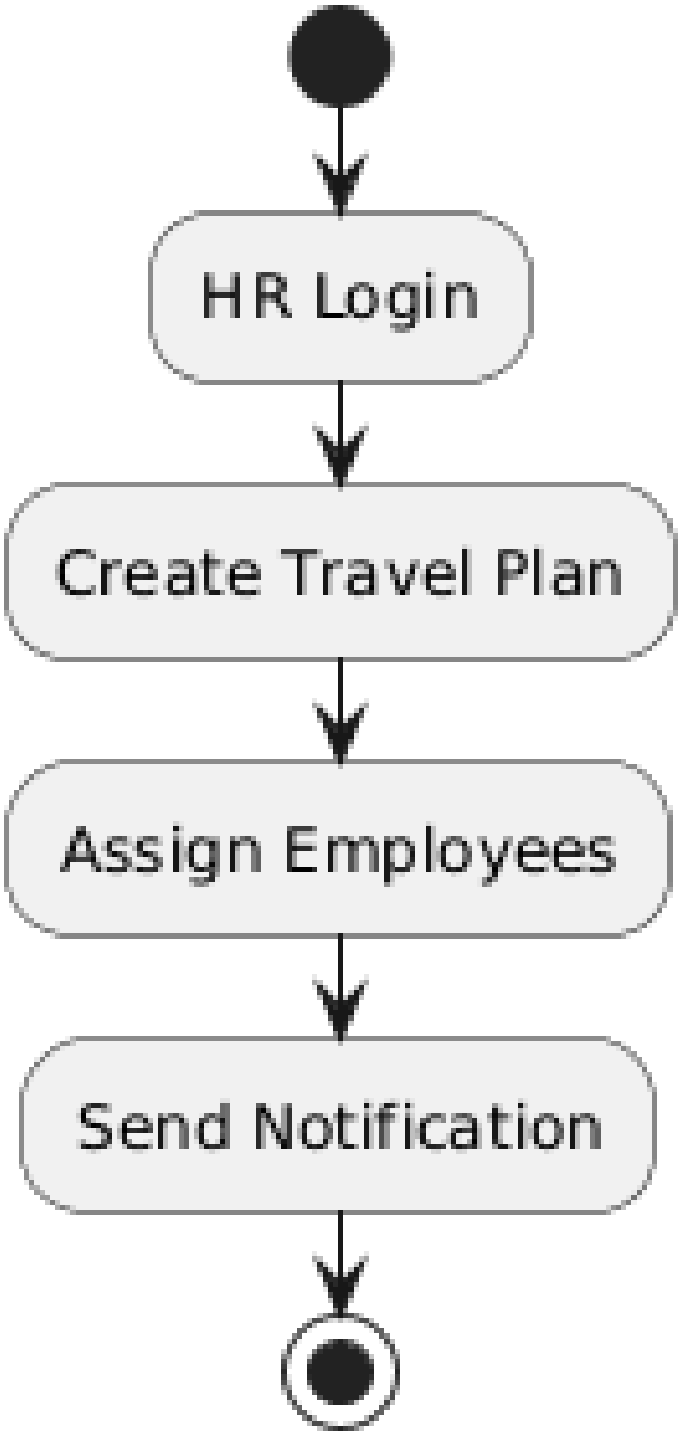


7.1.1.3 Employee usecase Diagram:

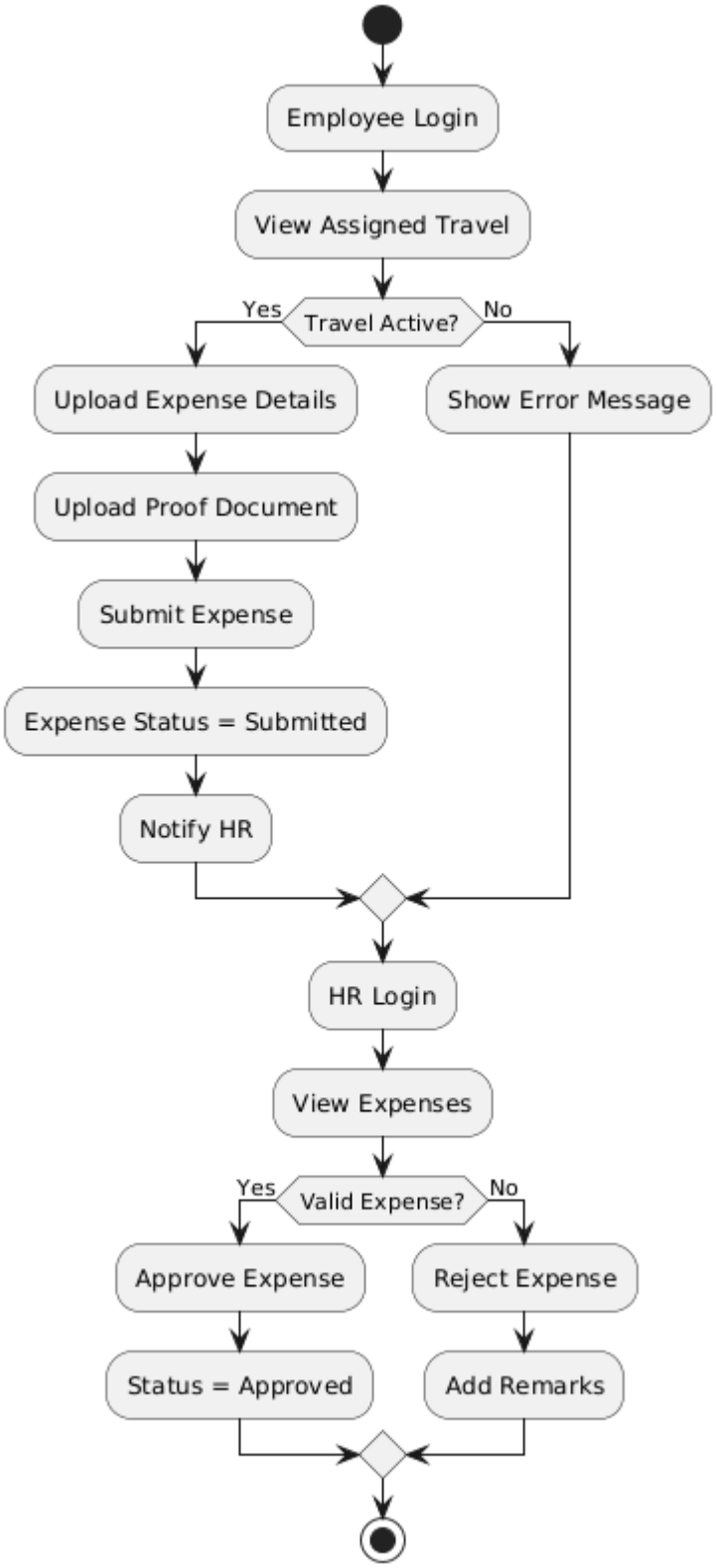


7.1.2 Activity Diagrams :

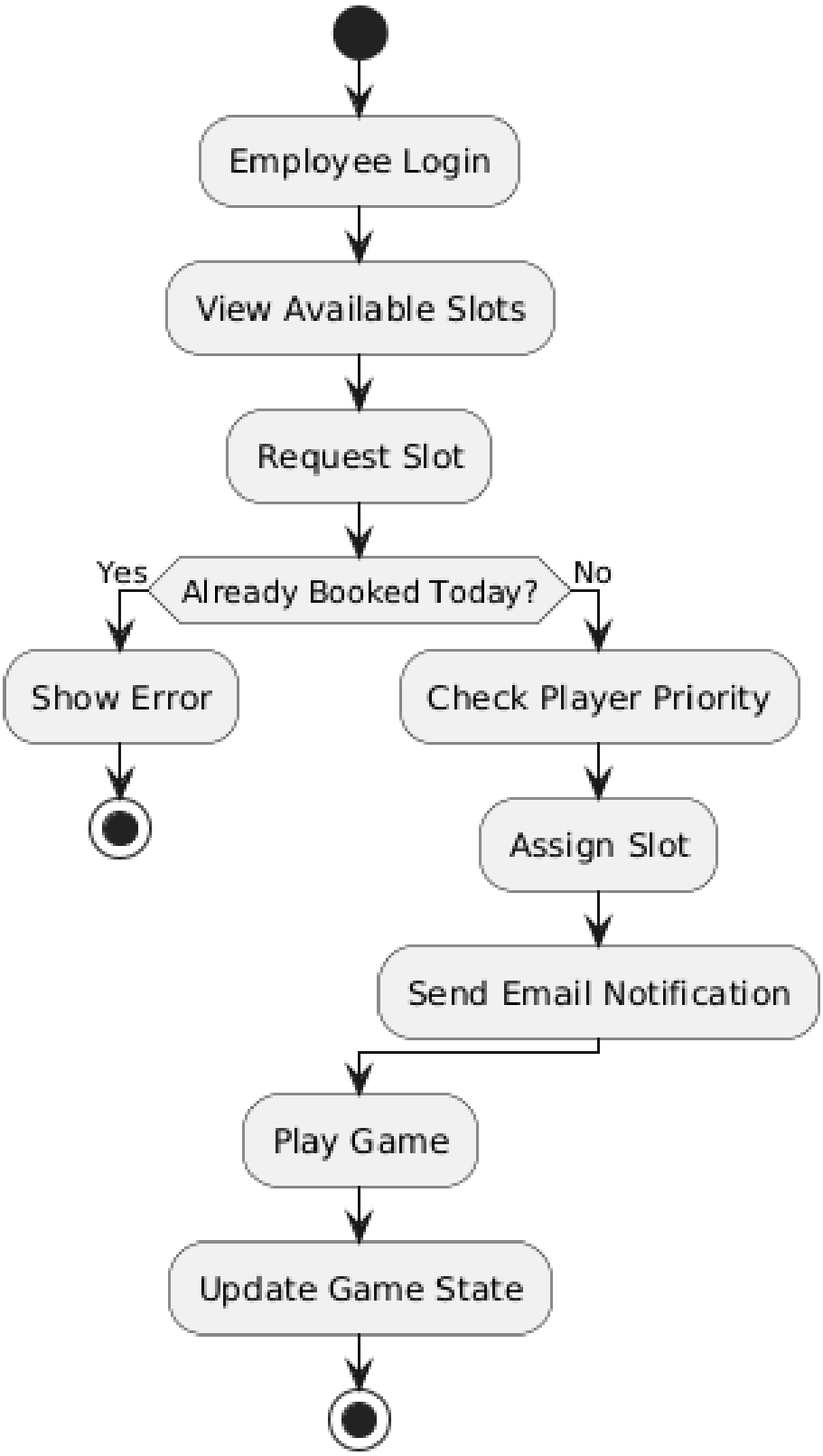
7.1.2.1 Create Travel :



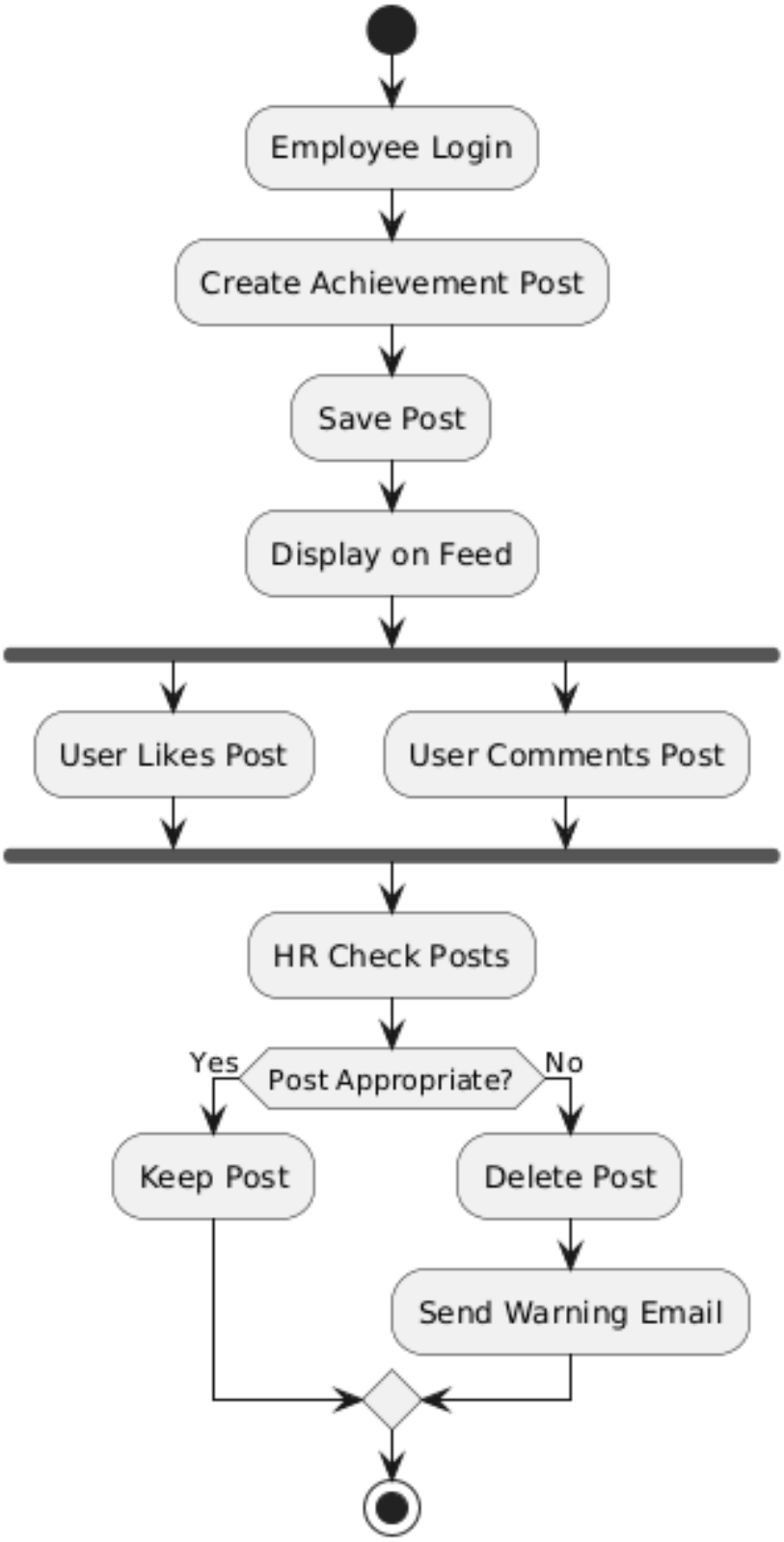
7.1.2.2 Travel Expense :



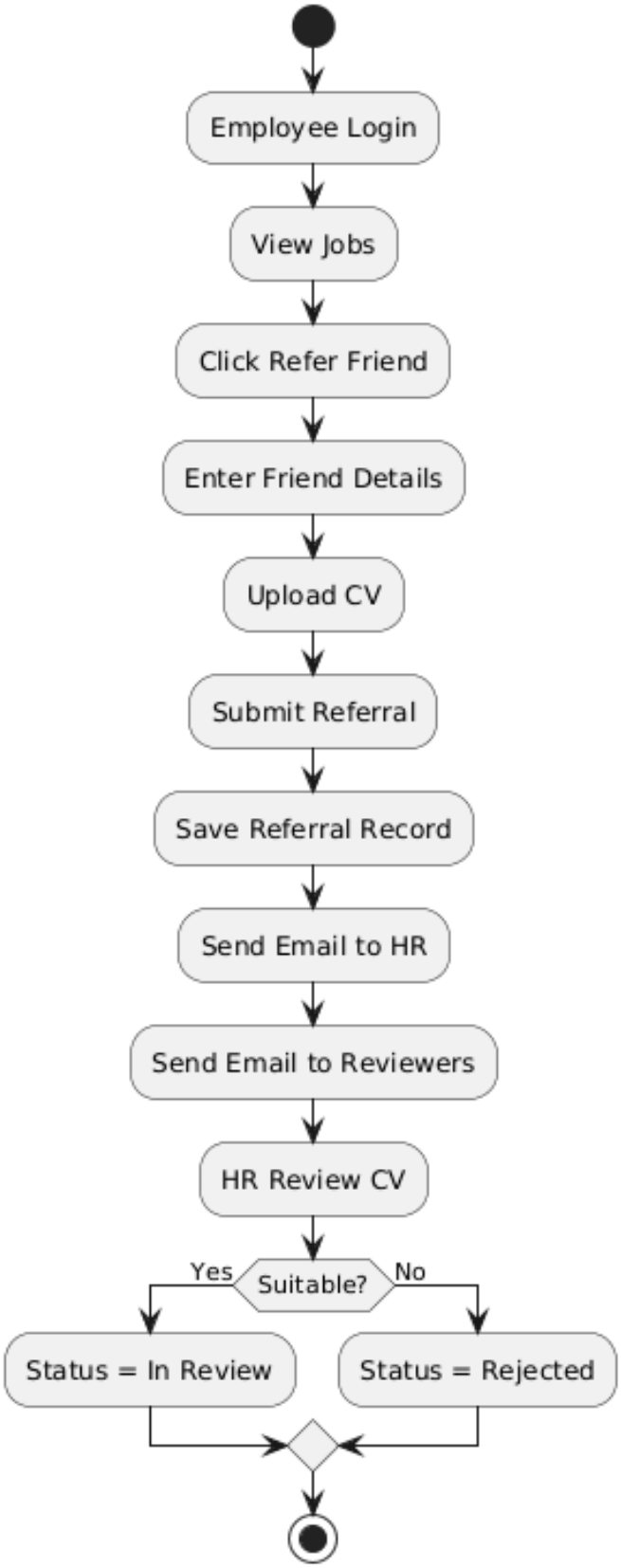
7.1.2.3 Book Game :



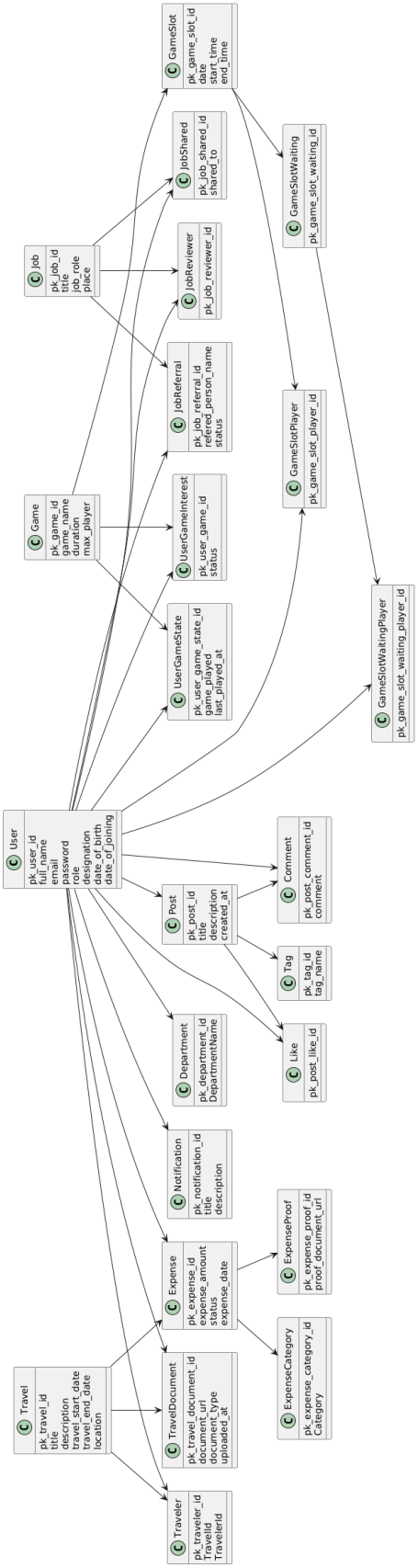
7.1.2.4 Post Create :



7.1.2.5 Job Referral :

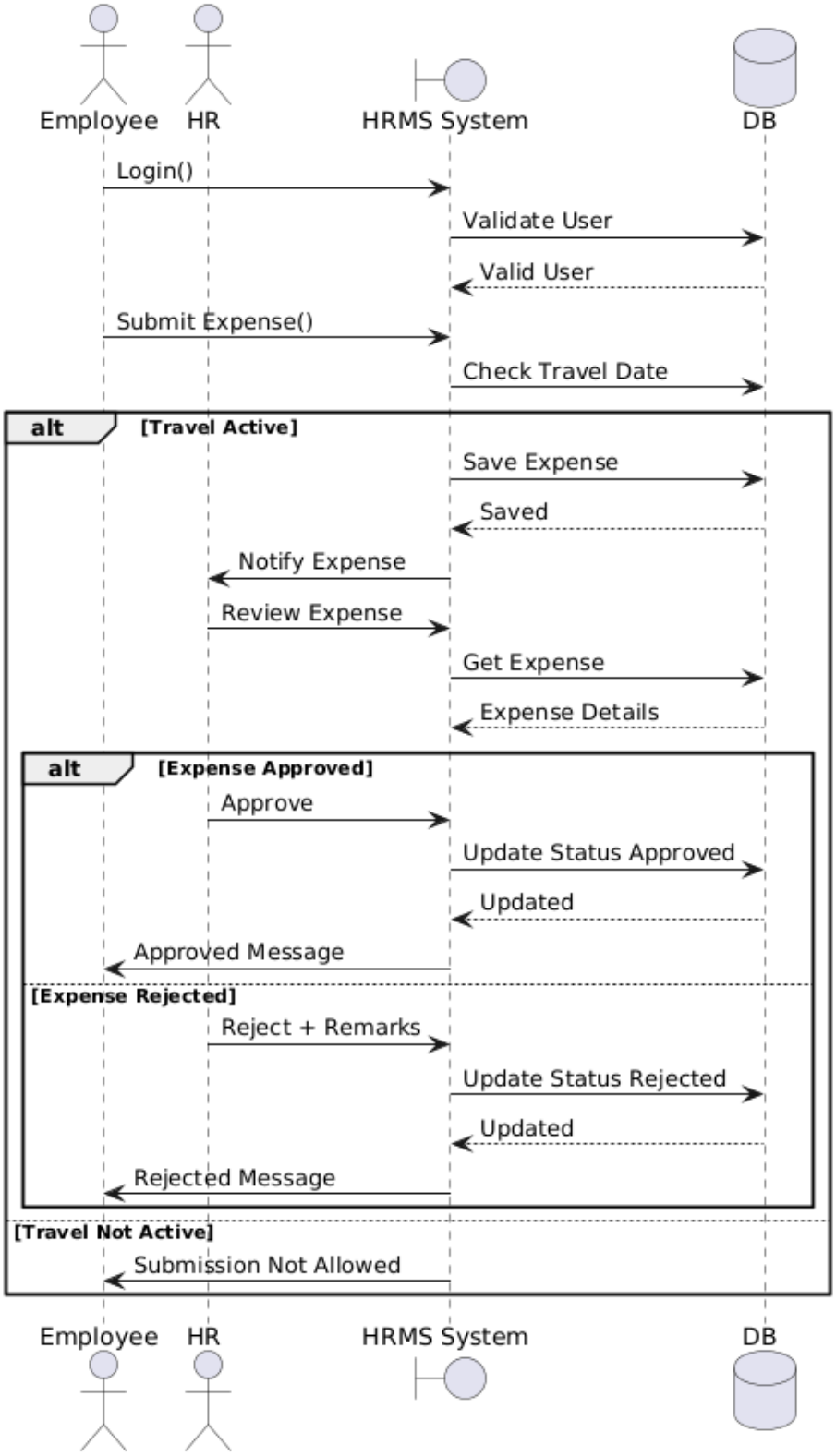


7.1.3 Class Diagram :

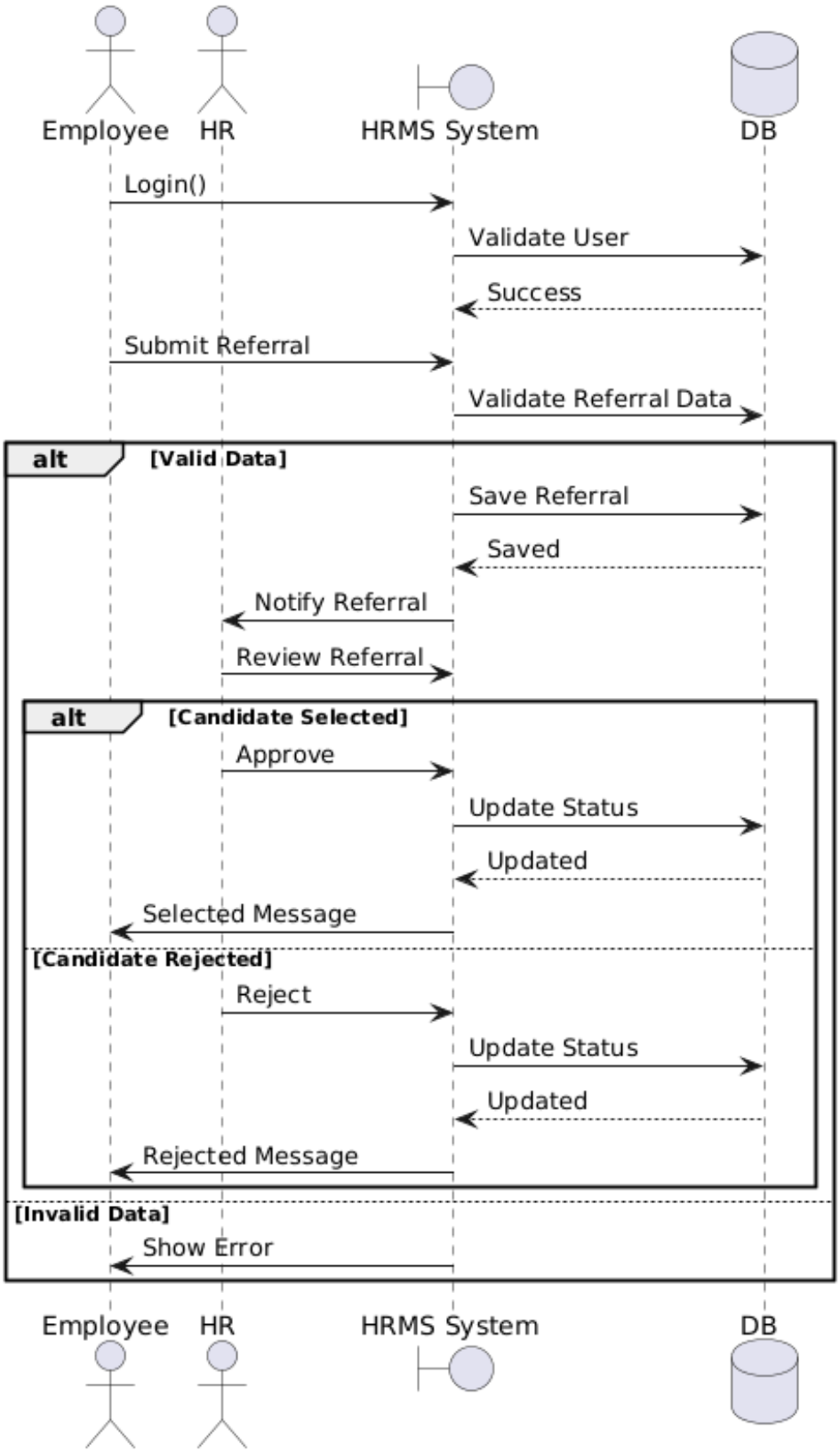


7.1.4 Sequence Diagram :

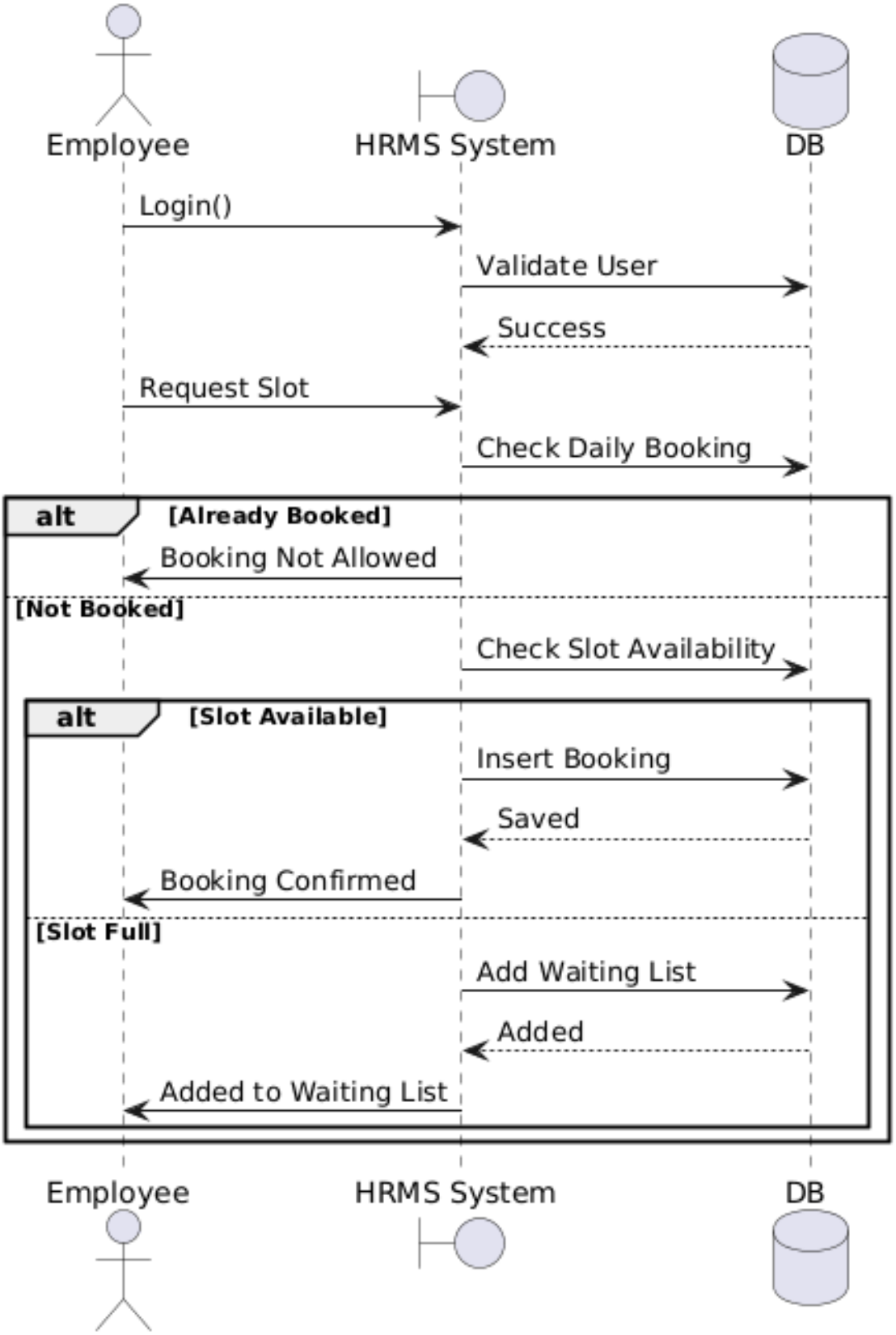
7.1.4.1 Travel Expense Sequence :



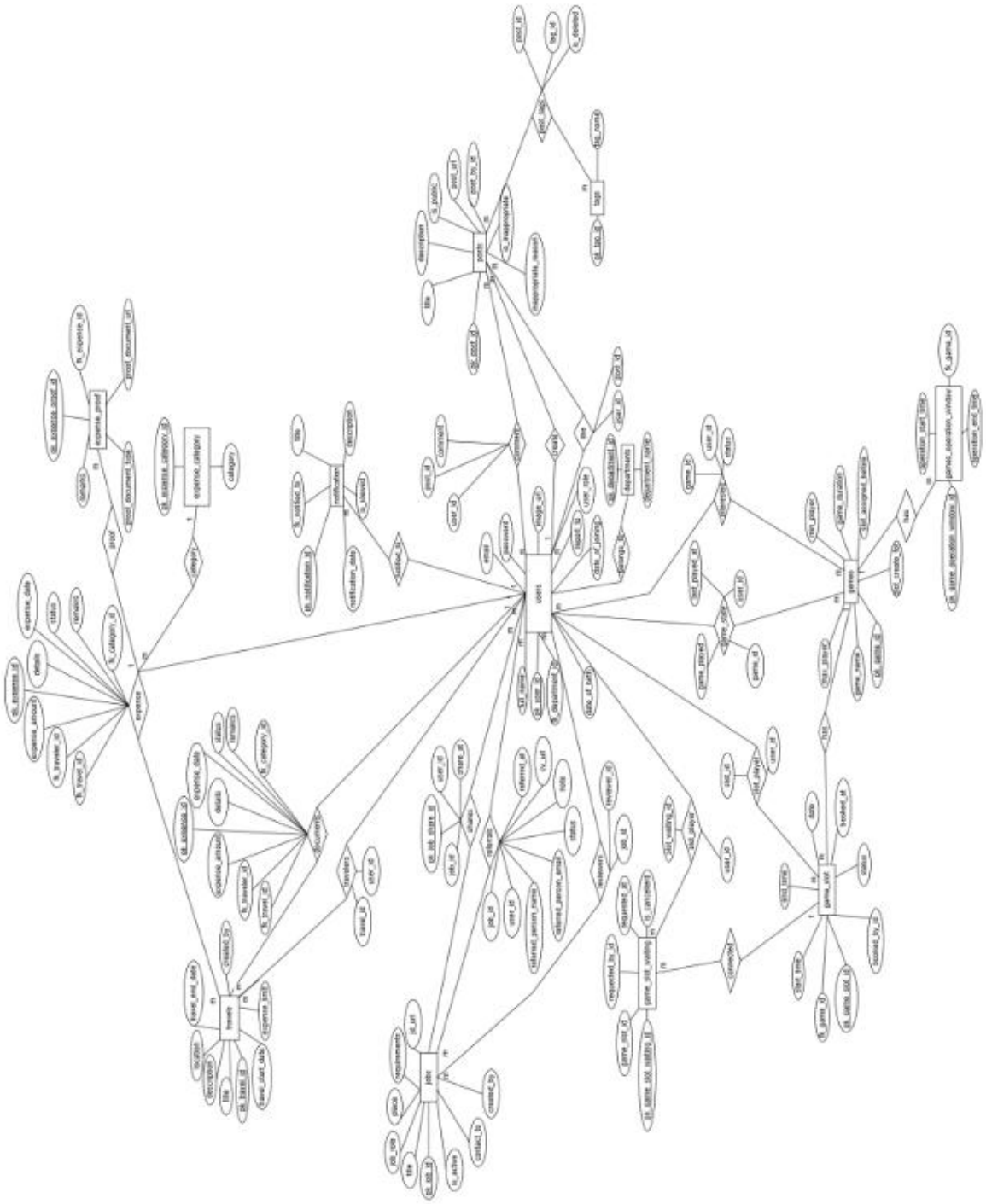
7.1.4.2 Job Referral Sequence :



7.1.4.3 Slot Booking Sequence :



7.2 ER Diagram :



7.3 Data Dictionary :

7.3.1 daily_celebrations : used to store daily celebration related data

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_daily_celebration_id	INT	-	PK, Identity	Celebration ID
2	UserId	INT	-	FK(users)	Employee
3	celebration_type	INT	-	NOT NULL	Birthday / Anniversary
4	celebration_date	DATETIME2	-	NOT NULL	Celebration date

7.3.2 departments : used to store department of system

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_department_id	INT	-	PK, Identity	Department ID
2	DepartmentName	NVARCHAR	30	NOT NULL	Department name
3	is_deleted	BIT	-	NOT NULL	Soft delete
4	created_at	DATETIME2	-	DEFAULT	Created time
5	updated_at	DATETIME2	-	DEFAULT	Updated time

7.3.3 expense_category : used to store expense category

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_expense_category_id	INT	-	PK, Identity	Category ID
2	Category	NVARCHAR	50	NOT NULL	Expense category
3	is_deleted	BIT	-	NOT NULL	Soft delete

7.3.4 **expense_proof :** used to store proof(document) related to expense

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_expense_proof_id	INT	-	PK, Identity	Proof ID
2	ExpenseId	INT	-	FK(expenses)	Expense reference
3	proof_document_url	NVARCHAR	500	NOT NULL	Document URL
4	expense_document_type	NVARCHAR	50	NOT NULL	Document type
5	remarks	NVARCHAR	200	NOT NULL	Remarks

7.3.5 **expense :** used to store expense related data

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_expense_id	INT	-	PK, Identity	Expense ID
2	TravelId	INT	-	FK(travels)	Travel
3	TravelerId	INT	-	FK(users)	Employee
4	expense_amount	DECIMAL	10,2	NOT NULL	Amount
5	CategoryId	INT	-	FK(expense_category)	Category
6	status	NVARCHAR	20	NOT NULL	Draft/Submitted/ Approved/Rejected
7	remarks	NVARCHAR	100	NOT NULL	Remarks
8	details	NVARCHAR	200	NOT NULL	Details
9	expense_date	DATETIME2	-	NOT NULL	Expense date
10	is_deleted	BIT	-	NOT NULL	Soft delete
11	created_at	DATETIME2	-	DEFAULT	Created time
12	updated_at	DATETIME2	-	DEFAULT	Updated time

7.3.6 **game_operation_window :** used to store operationg hours of game

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_game_operation_window_id	INT	-	PK, Identity	Window ID
2	GameId	INT	-	FK(games)	Game
3	operational_start_time	TIME	-	NOT NULL	Start time
4	operational_end_time	TIME	-	NOT NULL	End time
5	is_deleted	BIT	-	NOT NULL	Soft delete
6	created_at	DATETIME2	-	NOT NULL	Created time
7	updated_at	DATETIME2	-	NOT NULL	Updated time

7.3.7 **game_slot_players :** used to store players of the slot of game

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_game_slot_player_id	INT	-	PK, Identity	Game slot player ID
2	SlotId	INT	-	FK(game_slots)	Game slot
3	PlayerId	INT	-	FK(users)	Player

7.3.8 **game_slot_waiting :** used to store waiting list of the game

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_game_slot_waiting_id	INT	-	PK, Identity	Waiting ID
2	GameSlotId	INT	-	FK(game_slots)	Game slot
3	RequestedById	INT	-	FK(users)	Requested by
4	requested_at	DATETIME2	-	DEFAULT	Requested time
5	is_cancelled	BIT	-	NOT NULL	Cancellation flag

7.3.9 **game_slot_waiting_players :** used to store player listing who are in waiting list of specific game

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_game_slot_waiting_player_id	INT	-	PK, Identity	Waiting player ID
2	GameSlotWaitingId	INT	-	FK(game_slot_waiting)	Waiting slot
3	PlayerId	INT	-	FK(users)	Player

7.3.10 **game_slot :** used to store auto-generated slot of the game.

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_game_slot_id	INT	-	PK, Identity	Slot ID
2	GameId	INT	-	FK(games)	Game
3	start_time	TIME	-	NOT NULL	Start time
4	end_time	TIME	-	NOT NULL	End time
5	date	DATETIME2	-	NOT NULL	Slot date
6	BookedById	INT	-	FK(users)	Booked by
7	booked_at	DATETIME2	-	NULL	Booked time
8	Status	INT	-	NOT NULL	Slot status

7.3.11 **games :** used to store game of the system

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_game_id	INT	-	PK, Identity	Game ID
2	game_name	NVARCHAR	30	NOT NULL	Game name
3	max_player_per_game	INT	-	NOT NULL	Max players
4	min_player_per_game	INT	-	NOT NULL	Min players
5	duration_in_minutes	INT	-	NOT NULL	Duration
6	slot_assigned_before_x_minutes	INT	-	NOT NULL	Buffer time
7	slot_create_for_next_x_days	INT	-	NOT NULL	Slot generation days
8	is_deleted	BIT	-	NOT NULL	Soft delete
9	created_at	DATETIME2	-	DEFAULT	Created time
10	updated_at	DATETIME2	-	DEFAULT	Updated time

7.3.12 **job_referrals :** used to store job referrals on specific job

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_job_referral_id	INT	-	PK, Identity	Referral ID
2	refered_person_name	NVARCHAR	50	NOT NULL	Candidate name
3	refered_person_email	NVARCHAR	50	NOT NULL	Candidate email
4	cv_url	NVARCHAR	500	NOT NULL	CV URL
5	note	NVARCHAR	500	NOT NULL	Note
6	status	NVARCHAR	20	NOT NULL	Status
7	ReferedBy	INT	-	FK(users)	Referred by
8	JobId	INT	-	FK(jobs)	Job
9	is_deleted	BIT	-	NOT NULL	Soft delete
10	refered_at	DATETIME2	-	NOT NULL	Referral date

7.3.13 **job_reviewers :** used to store reviewers of specific job

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_job_reviewer_id	INT	-	PK, Identity	Job reviewer ID
2	JobId	INT	-	FK(jobs)	Job
3	ReviewerId	INT	-	FK(users)	Reviewer

7.3.14 **job_shared :** used to store job share records

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_job_shared_id	INT	-	PK, Identity	Job share ID
2	shared_to	NVARCHAR	50	NOT NULL	Shared email
3	shared_at	INT	-	FK(users)	Shared by
4	JobId	INT	-	FK(jobs)	Job
5	is_deleted	BIT	-	NOT NULL	Soft delete
6	SharedAt	DATETIME2	-	NOT NULL	Shared date

7.3.15 **jobs :** used to store jobs of the system

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_job_id	INT	-	PK, Identity	Job ID
2	title	NVARCHAR	50	NOT NULL	Job title
3	job_role	NVARCHAR	20	NOT NULL	Role
4	place	NVARCHAR	30	NOT NULL	Location
5	requirements	NVARCHAR	300	NOT NULL	Requirements
6	jd_url	NVARCHAR	500	NOT NULL	JD URL
7	CreatedBy	INT	-	FK(users)	Created by
8	ContactTo	INT	-	FK(users)	HR contact
9	is_active	BIT	-	NOT NULL	Active
10	is_deleted	BIT	-	NOT NULL	Soft delete
11	created_at	DATETIME2	-	DEFAULT	Created time
12	updated_at	DATETIME2	-	DEFAULT	Updated time

7.3.16 **notifications :** used to store notification of system

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_notification_id	INT	-	PK, Identity	Notification ID
2	NotifiedTo	INT	-	FK(users)	User
3	title	NVARCHAR	50	NOT NULL	Title
4	description	NVARCHAR	500	NOT NULL	Description
5	is_viewed	BIT	-	NOT NULL	Viewed
6	notification_date	DATETIME2	-	NOT NULL	Date

7.3.17 **post_comments :** used to store comment on posts.

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_post_comment_id	INT	-	PK, Identity	Comment ID
2	comment	NVARCHAR	500	NOT NULL	Comment text
3	PostId	INT	-	FK(posts)	Post
4	CommentById	INT	-	FK(users)	Commented by
5	is_deleted	BIT	-	NOT NULL	Soft delete
6	created_at	DATETIME2	-	DEFAULT	Created time
7	updated_at	DATETIME2	-	DEFAULT	Updated time

7.3.18 **post_likes :** used to store like on the post

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_post_like_id	INT	-	PK, Identity	Like ID
2	PostId	INT	-	FK(posts)	Post
3	LikedById	INT	-	FK(users)	Liked by
4	is_deleted	BIT	-	NOT NULL	Soft delete

7.3.19 **post_tages :** used to store tag of the post

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_post_tag_id	INT	-	PK, Identity	Post tag ID
2	PostId	INT	-	FK(posts)	Post
3	TagId	INT	-	FK(tags)	Tag
4	is_deleted	BIT	-	NOT NULL	Soft delete

7.3.20 **posts :** used to store post created by users

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_post_id	INT	-	PK, Identity	Post ID
2	post_url	NVARCHAR	500	NOT NULL	Media URL
3	PostById	INT	-	FK(users)	Posted by
4	title	NVARCHAR	100	NOT NULL	Title
5	description	NVARCHAR	1000	NOT NULL	Description
6	is_public	BIT	-	NOT NULL	Visibility
7	is_inappropriate	BIT	-	NOT NULL	Moderation flag
8	inappropriate_reason	NVARCHAR	500	NULL	Reason
9	is_deleted	BIT	-	NOT NULL	Soft delete
10	created_at	DATETIME2	-	DEFAULT	Created
11	updated_at	DATETIME2	-	DEFAULT	Updated

7.3.21 **tags :** used to store common tag of system for post

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_tag_id	INT	-	PK, Identity	Tag ID
2	tag_name	NVARCHAR	30	NOT NULL	Tag name
3	IsDeleted	BIT	-	NOT NULL	Soft delete

7.3.22 **travel_documents :** used to store document of travel

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_travel_document_id	INT	-	PK, Identity	Document ID
2	TravelId	INT	-	FK(travels)	Travel
3	TravelerId	INT	-	FK(users)	Traveler
4	document_url	NVARCHAR	400	NOT NULL	Document URL
5	document_name	NVARCHAR	30	NOT NULL	Document name
6	document_type	NVARCHAR	50	NOT NULL	Document type
7	UploadedBy	INT	-	FK(users)	Uploaded by
8	uploaded_at	DATETIME2	-	NOT NULL	Upload time
9	is_deleted	BIT	-	NOT NULL	Soft delete

7.3.23 **travellers :** used to store travellers of specific travel

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_traveler_id	INT	-	PK, Identity	Traveler ID
2	TravelId	INT	-	FK(travels)	Travel
3	TravelerId	INT	-	FK(users)	Traveler
4	is_deletd	BIT	-	NOT NULL	Soft delete

7.3.24 **travels :** used to store travel data

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_travel_id	INT	-	PK, Identity	Travel ID
2	title	NVARCHAR	50	NOT NULL	Title
3	description	NVARCHAR	300	NOT NULL	Description
4	travel_start_date	DATETIME2	-	NOT NULL	Start date
5	trael_end_date	DATETIME2	-	NOT NULL	End date
6	location	NVARCHAR	50	NOT NULL	Location
7	CreatedBy	INT	-	FK(users)	Created by
8	expense_max_amount_limit	DECIMAL	10,2	NOT NULL	Limit
9	is_deleted	BIT	-	NOT NULL	Soft delete
10	created_at	DATETIME2	-	DEFAULT	Created
11	updated_at	DATETIME2	-	DEFAULT	Updated

7.3.25 **user_game_interest :** used to store game interest of user in specific game

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_user_game_id	INT	-	PK, Identity	Interest ID
2	GameId	INT	-	FK(games)	Game
3	UserId	INT	-	FK(users)	User
4	status	NVARCHAR	20	NOT NULL	Interest status

7.3.26 **user_game_state :** used to store state of user on specific game

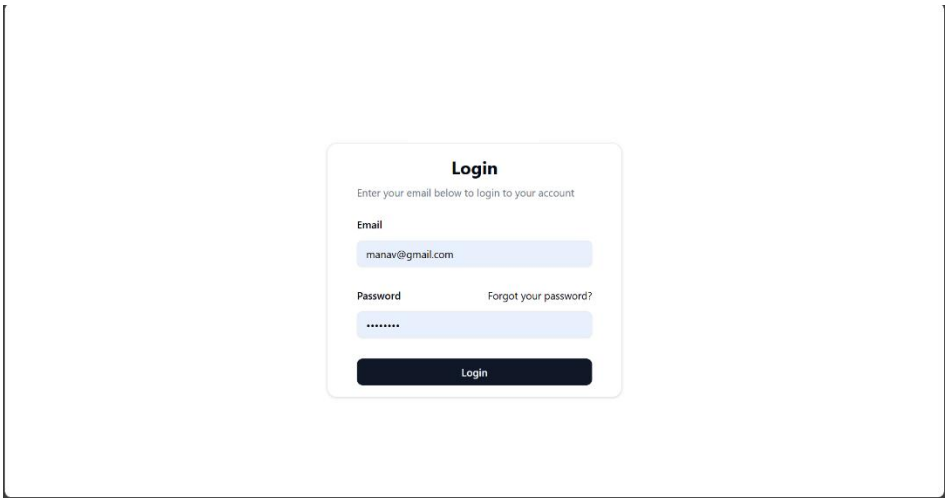
Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_user_game_state_id	INT	-	PK, Identity	Game state ID
2	GameId	INT	-	FK(games)	Game
3	UserId	INT	-	FK(users)	User
4	game_played	INT	-	NOT NULL	Times played
5	last_played_at	DATETIME2	-	NOT NULL	Last played time

7.3.27 **users :** used to store data related to system user

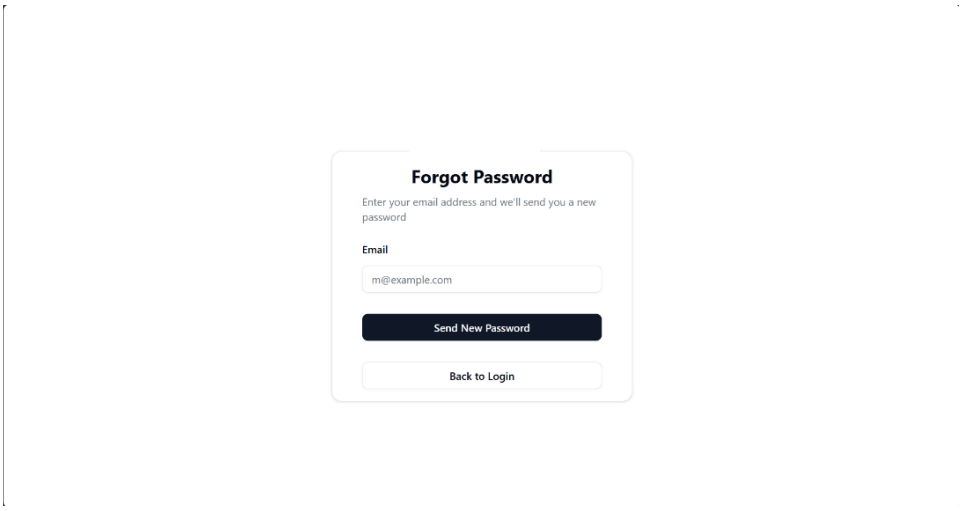
Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_user_id	INT	-	PK, Identity	User ID
2	full_name	NVARCHAR	150	NOT NULL	Full name
3	email	NVARCHAR	50	UNIQUE, NOT NULL	Email
4	password	NVARCHAR	500	NOT NULL	Password
5	image_url	NVARCHAR	500	NOT NULL	Profile image
6	user_role	NVARCHAR	15	NOT NULL	Role
7	date_of_birth	DATETIME2	-	NOT NULL	DOB
8	date_of_joining	DATETIME2	-	NOT NULL	DOJ
9	ReportTo	INT	-	FK(users)	Manager
10	DepartmentId	INT	-	FK(departments)	Department
11	designation	NVARCHAR	30	NOT NULL	Designation
12	is_deleted	BIT	-	NOT NULL	Soft delete
13	created_at	DATETIME2	-	DEFAULT	Created
14	updated_at	DATETIME2	-	DEFAULT	Updated

8. IMPLEMENTATION OF SYSTEM

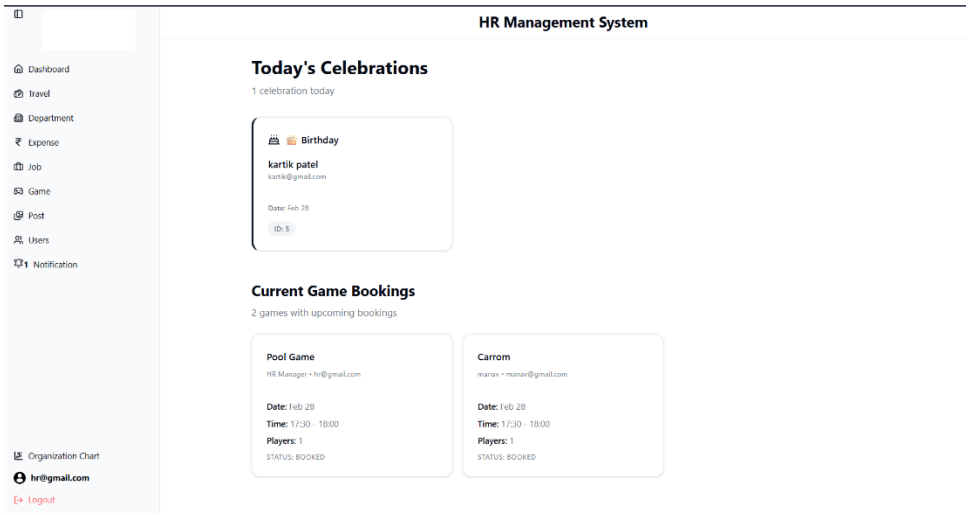
8.1 Screen Snapshots :



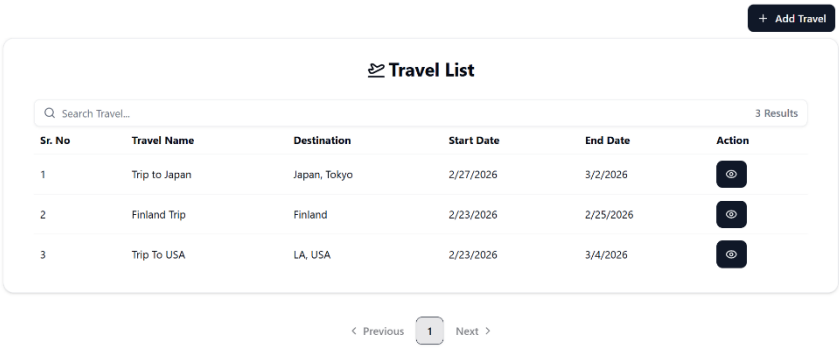
- Login Page Used to login into system



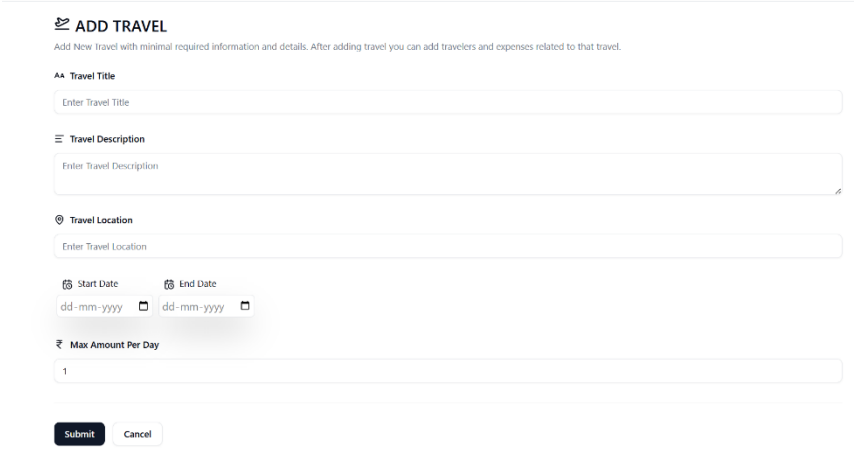
- Forget password to generate new password and get in email



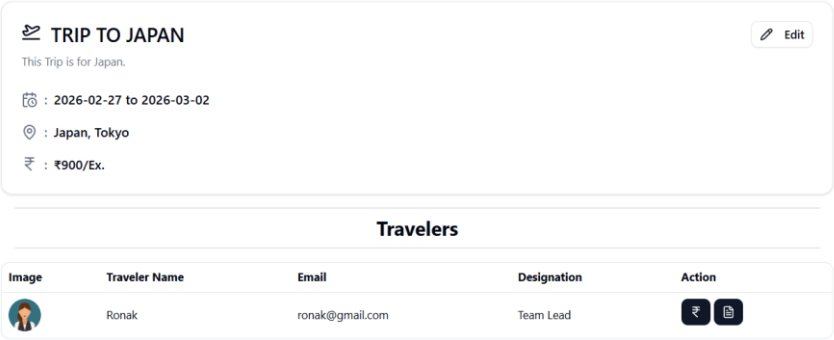
- Dashboard for HR of system



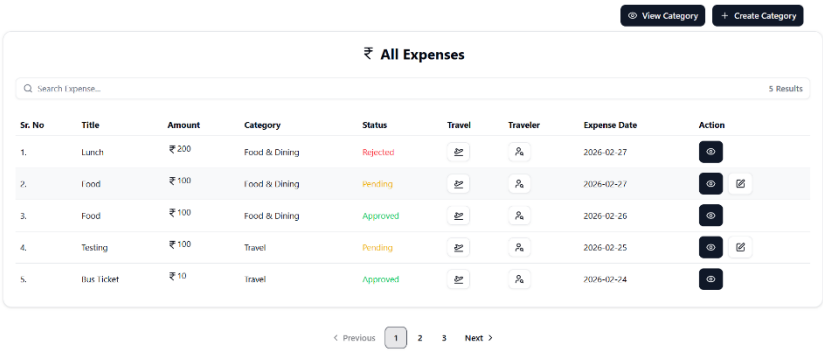
- List All Travel Created by Logged in HR



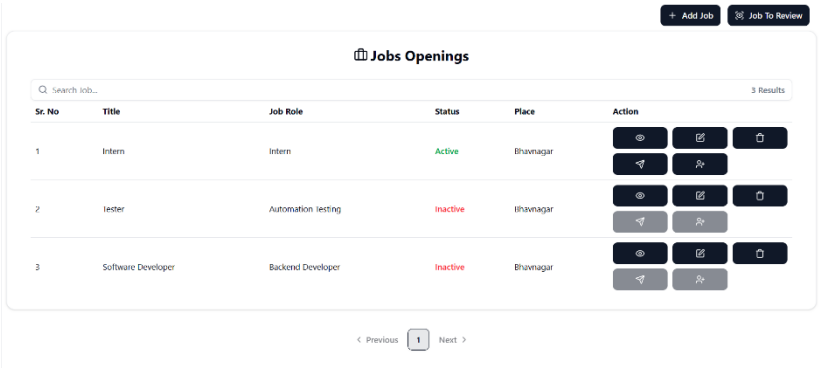
- Travel Create Page



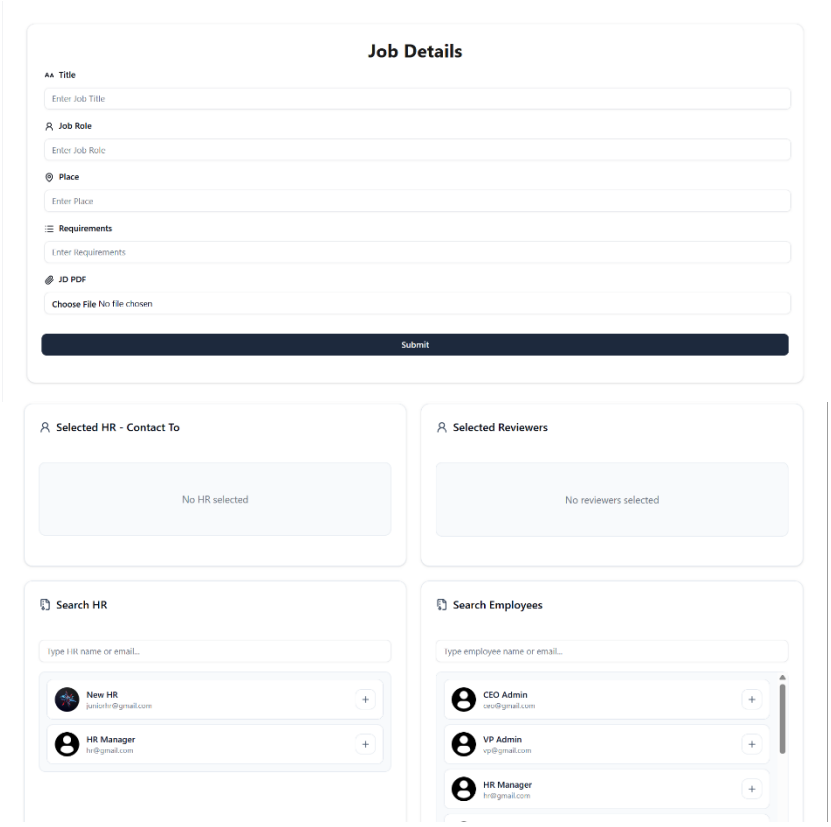
- Travel Detail Page with travellers



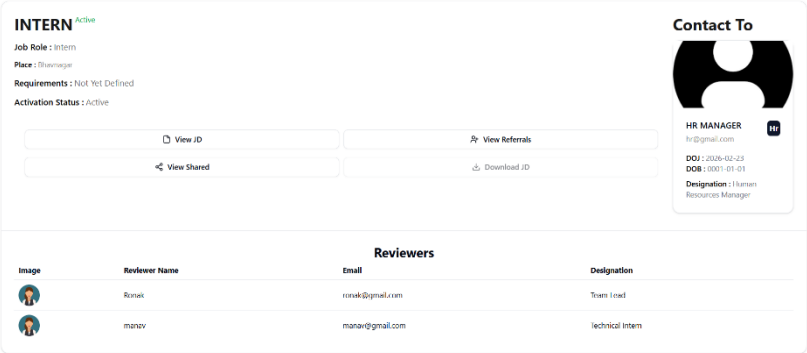
- All Travel Expenses of Traveller for changing status of it and preview proof of expense.



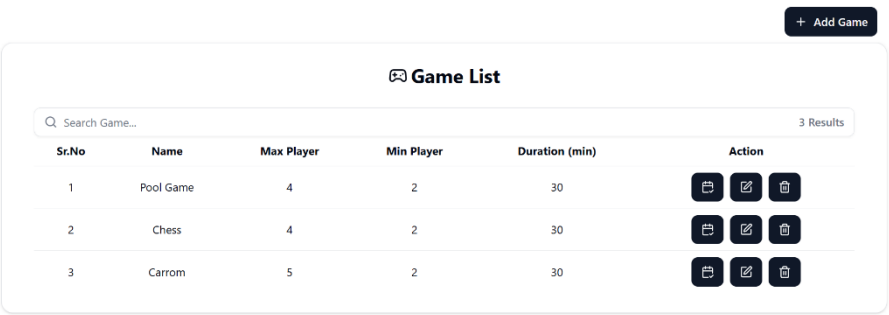
- Job Opening Created by logged in HR.



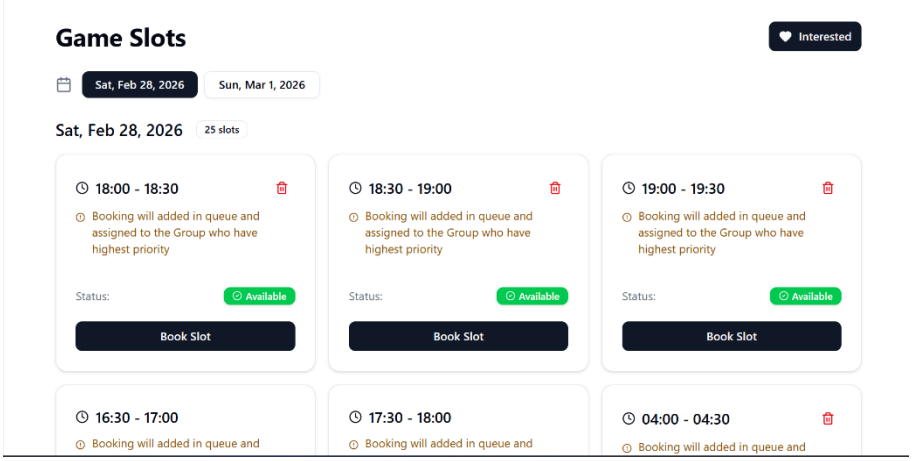
- Create Job For Opening.



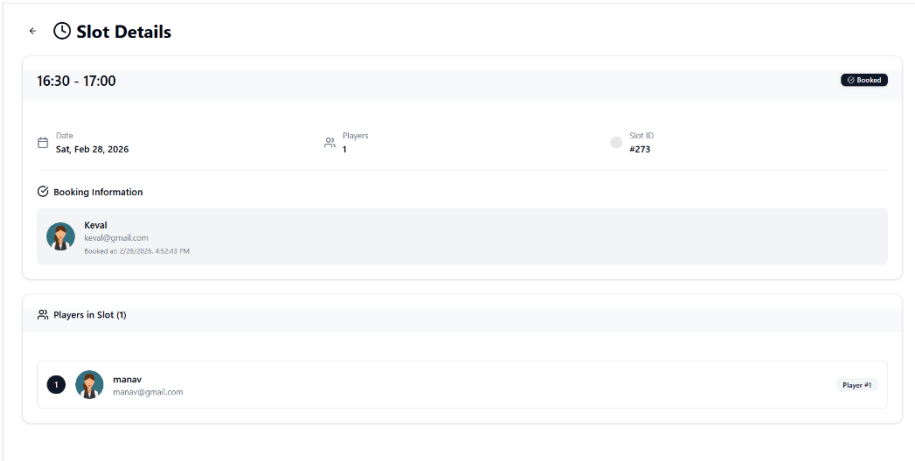
- Job Detail with reviewers and Contact to Detail



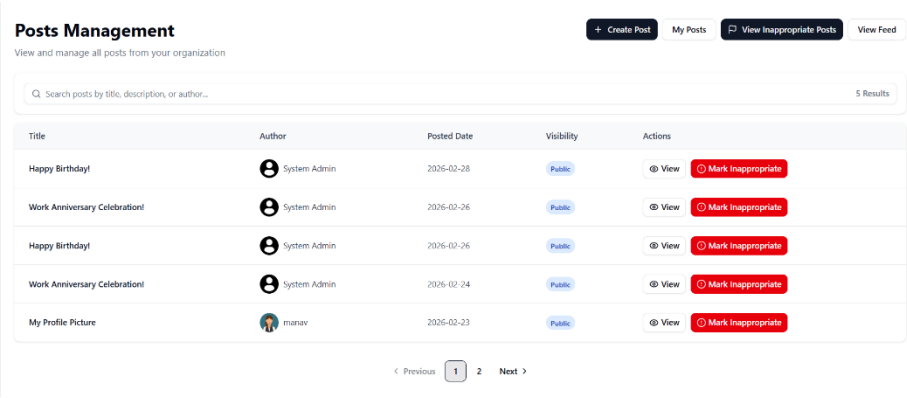
- All Game Of System.



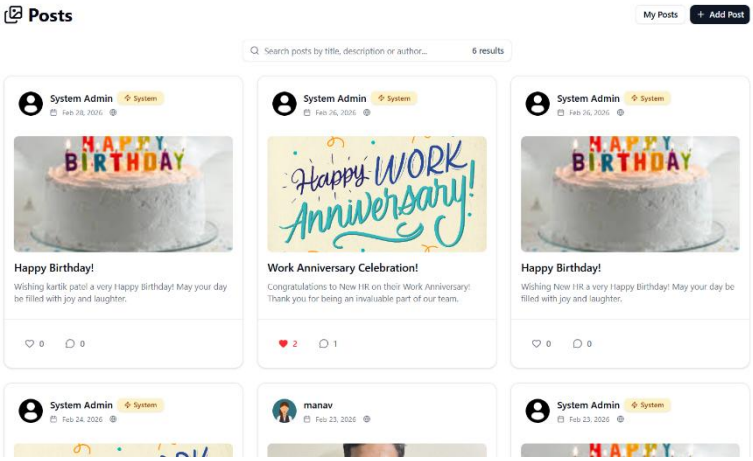
- Game Slot



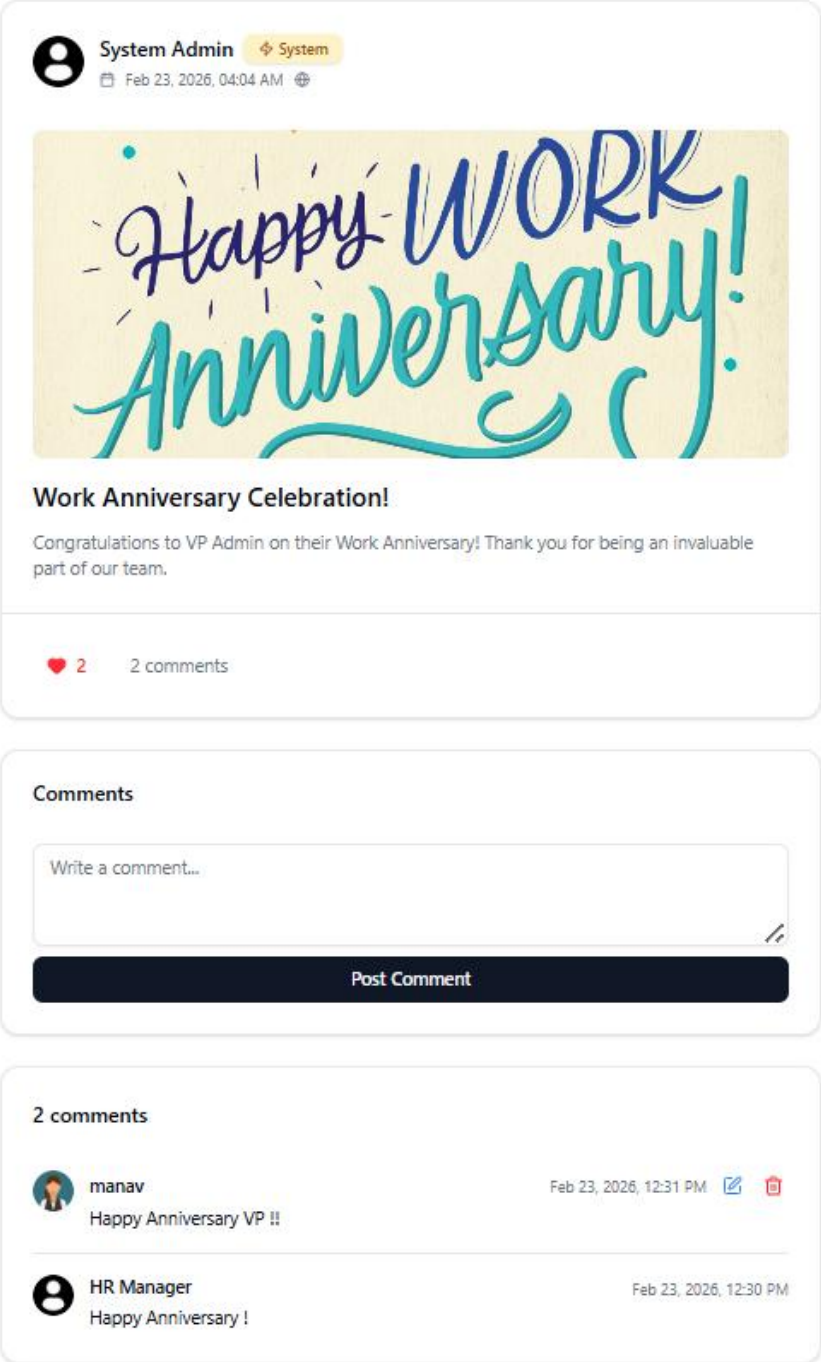
- Game Slot Booking Details with who booked and player of that slot.



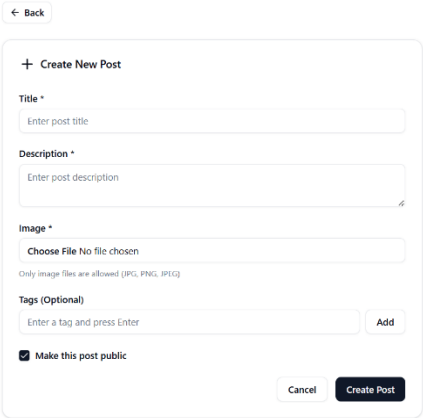
- Manage Post By HR to make post inappropriate or view details.



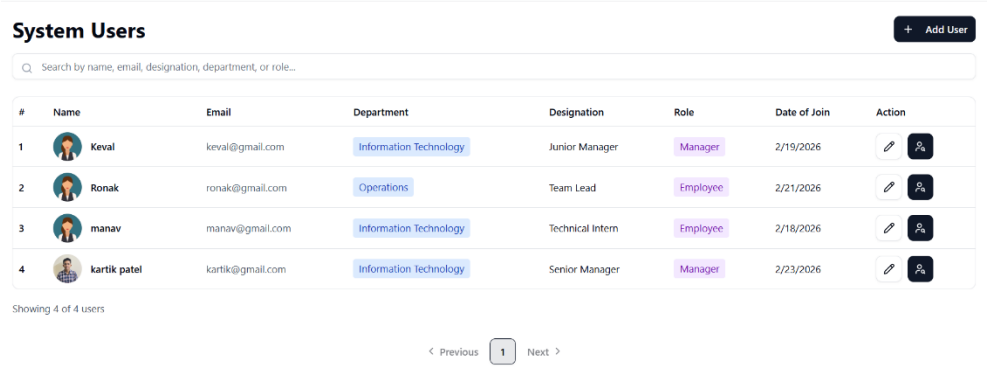
- Post Feed For All User who are logged in system



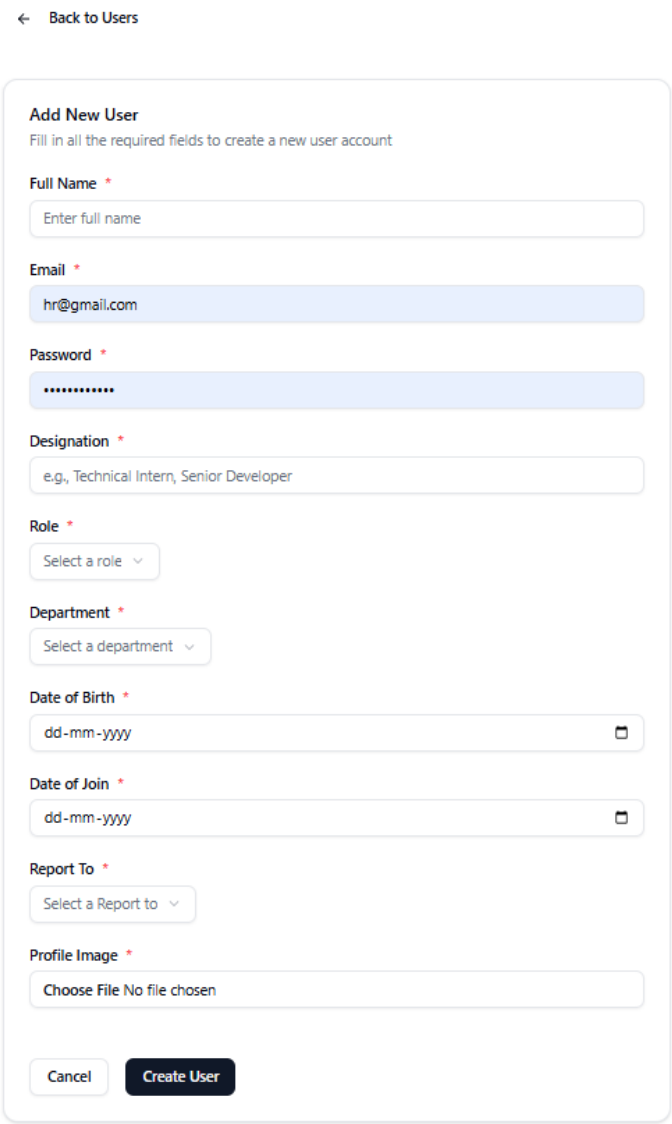
- Post Detail with Comment and like count.



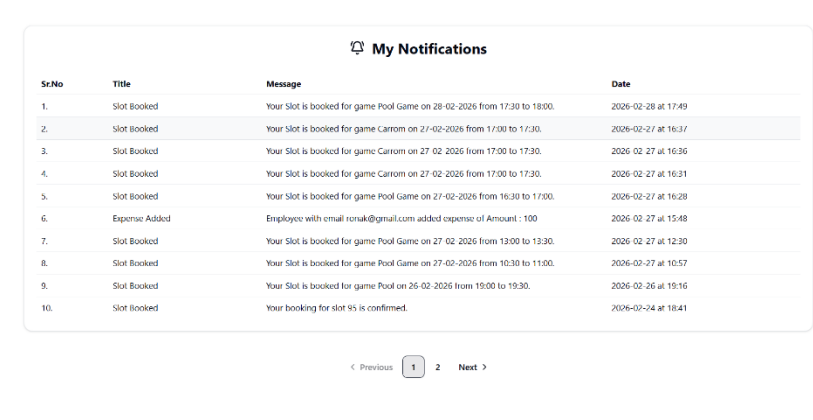
- Create Post Form.



- Manage User Page for HR



- Create User Page.

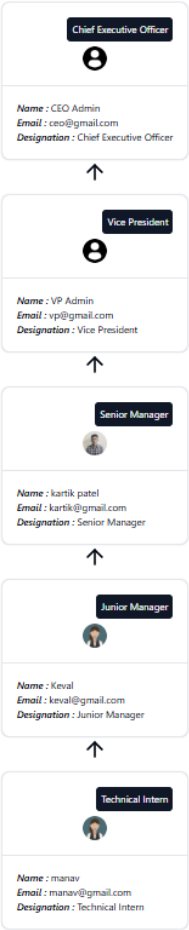


My Notifications

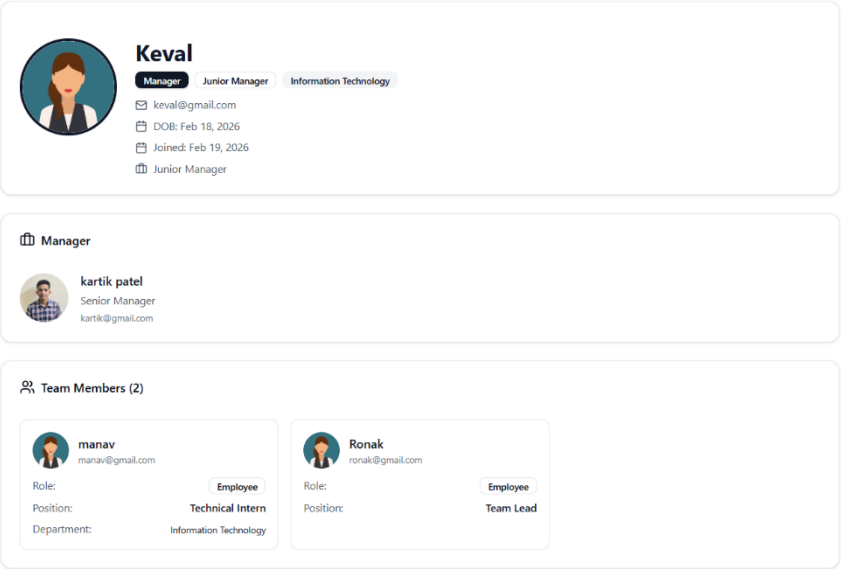
Sr.No	Title	Message	Date
1.	Slot Booked	Your Slot is booked for game Pool Game on 28-02-2026 from 17:30 to 18:00.	2026-02-28 at 17:49
2.	Slot Booked	Your Slot is booked for game Carrom on 27-02-2026 from 17:00 to 17:30.	2026-02-27 at 16:37
3.	Slot Booked	Your Slot is booked for game Carrom on 27-02-2026 from 17:00 to 17:30.	2026-02-27 at 16:36
4.	Slot Booked	Your Slot is booked for game Carrom on 27-02-2026 from 17:00 to 17:30.	2026-02-27 at 16:31
5.	Slot Booked	Your Slot is booked for game Pool Game on 27-02-2026 from 16:30 to 17:00.	2026-02-27 at 16:28
6.	Expense Added	Employee with email ronak@gmail.com added expense of Amount : 100	2026-02-27 at 15:48
7.	Slot Booked	Your Slot is booked for game Pool Game on 27-02-2026 from 13:00 to 13:30.	2026-02-27 at 12:30
8.	Slot Booked	Your Slot is booked for game Pool Game on 27-02-2026 from 10:30 to 11:00.	2026-02-27 at 10:57
9.	Slot Booked	Your Slot is booked for game Pool on 26-02-2026 from 19:00 to 19:30.	2026-02-26 at 19:16
10.	Slot Booked	Your booking for slot 85 is confirmed.	2026-02-24 at 18:41

Previous 1 2 Next

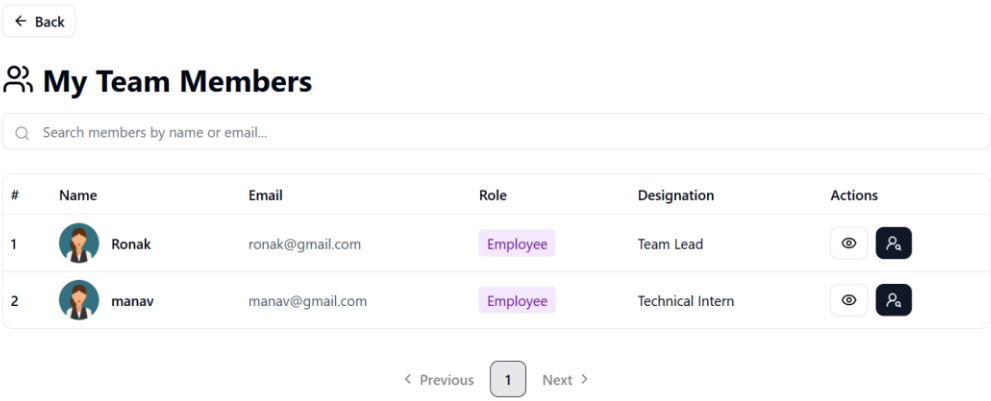
- Notification of current logged in user.



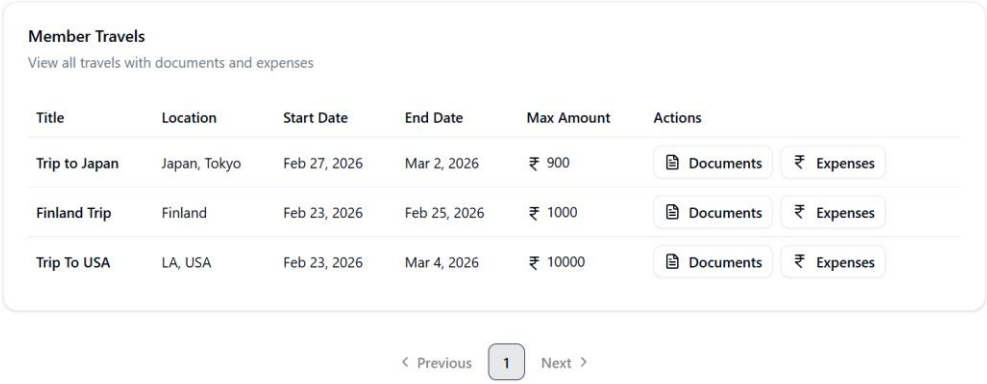
- Organization Chart



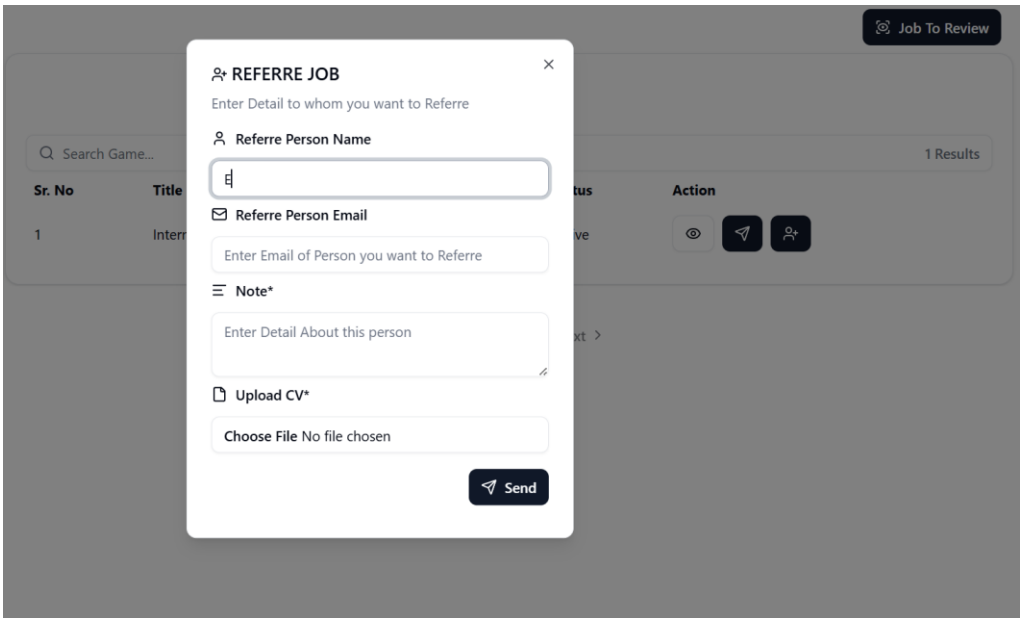
- User Profile with his Manager and His Team(if user is manager)



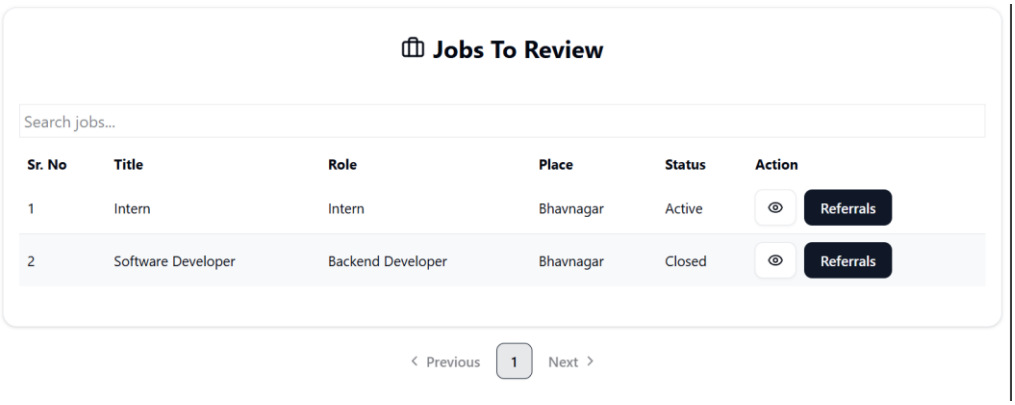
- My Team Detail for Manager



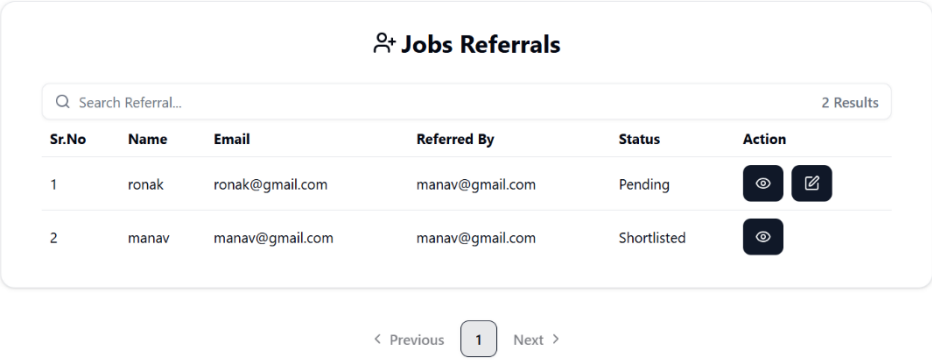
- Team Member Travel Detail



- Job Referral Create Form



- Job To review Page for Job Reviewers



- Job Referrals for Review.

8.2 Reports :

8.3 Testing :

8.3.1 Testing Strategy :

- **Black-Box Testing**

Black box testing is a testing technique in which the internal workings of the software are not known to the tester. The tester only focuses on the input and output of the software. Black box testing is mainly focused on testing the functionality of the software, ensuring that it meets the requirements and specifications. Black box testing uses methods like equivalence partitioning, boundary value analysis, and error guessing to create test cases.

8.3.2 Test cases and results :

1. **Add Travel Form:**

Test Case ID	Form/Screen	Field Name	Test Scenario	Input Data	Expected Result	Actual Result	Status (Pass / Fail)
1	Add Travel Form	Travel Title	Valid Travel Title	Business Trip	Accepted	Accepted	Pass
2	Add Travel Form	Travel Title	Empty Travel Title		Validation Error	Validation Error	Pass
3	Add Travel Form	Travel Title	Title exceeds maximum length	300 characters	Validation Error	Validation Error	Pass
4	Add Travel Form	Travel Description	Valid Description	Client meeting in Delhi office	Accepted	Accepted	Pass
5	Add Travel Form	Travel Description	Empty Description		Validation Error	Validation Error	Pass
6	Add Travel Form	Travel Location	Valid Location	Delhi	Accepted	Accepted	Pass
7	Add Travel Form	Travel Location	Empty Location		Validation Error	Validation Error	Pass
8	Add Travel Form	Start Date	Valid Start Date	10-06-2026	Accepted	Accepted	Pass
9	Add Travel Form	Start Date	Empty Start Date		Validation Error	Validation Error	Pass
10	Add Travel Form	End Date	Valid End Date	15-06-2026	Accepted	Accepted	Pass
11	Add Travel Form	End Date	End Date earlier than Start Date	Start: 15-06-2026 End: 10-06-2026	Validation Error	Validation Error	Pass
12	Add Travel Form	End Date	Empty End Date		Validation Error	Validation Error	Pass
13	Add Travel Form	Max Amount Per Day	Valid Amount	2000	Accepted	Accepted	Pass

Test Case ID	Form/Screen	Field Name	Test Scenario	Input Data	Expected Result	Actual Result	Status (Pass / Fail)
14	Add Travel Form	Max Amount Per Day	Empty Amount		Validation Error	Validation Error	Pass
15	Add Travel Form	Max Amount Per Day	Negative Amount	-500	Validation Error	Validation Error	Pass
16	Add Travel Form	Max Amount Per Day	Non-numeric input	abc	Validation Error	Validation Error	Pass
17	Add Travel Form	Submit Button	Submit with valid data	Valid Form Data	Travel record created successfully	Travel record created successfully	Pass
18	Add Travel Form	Cancel Button	Click cancel button	—	User redirected to travel list page	Redirect successful	Pass

2. Job Opening Create Form:

Test Case ID	Form/Screen	Field Name	Test Scenario	Input Data	Expected Result	Actual Result	Status (Pass / Fail)
1	Job Details Form	Title	Valid Job Title	Software Engineer	Accepted	Accepted	Pass
2	Job Details Form	Title	Empty Job Title		Validation Error	Validation Error	Pass
3	Job Details Form	Title	Very long title	300 characters	Validation Error	Validation Error	Pass
4	Job Details Form	Job Role	Valid Job Role	Backend Developer	Accepted	Accepted	Pass
5	Job Details Form	Job Role	Empty Job Role		Validation Error	Validation Error	Pass
6	Job Details Form	Place	Valid Place	Ahmedabad	Accepted	Accepted	Pass
7	Job Details Form	Place	Empty Place		Validation Error	Validation Error	Pass

Test Case ID	Form/Screen	Field Name	Test Scenario	Input Data	Expected Result	Actual Result	Status (Pass / Fail)
8	Job Details Form	Requirements	Valid Requirements	Java, Spring Boot, SQL	Accepted	Accepted	Pass
9	Job Details Form	Requirements	Empty Requirements		Validation Error	Validation Error	Pass
10	Job Details Form	JD PDF	Upload valid PDF file	JD.pdf	Accepted	Accepted	Pass
11	Job Details Form	JD PDF	Upload invalid file type	JD.png	Validation Error	Validation Error	Pass
12	Job Details Form	JD PDF	No file uploaded		Validation Error	Validation Error	Pass
13	HR Selection Panel	Search HR	Search HR by name	Komal Patel	HR appears in list	HR appears in list	Pass
14	HR Selection Panel	Search HR	Search HR by email	hr@gmail.com	HR appears in list	HR appears in list	Pass
15	HR Selection Panel	Select HR	Add HR using + button	Komal Patel	HR added to "Selected HR" section	HR added successfully	Pass
16	HR Selection Panel	Select HR	Select multiple HR users	Komal Patel, Falguni Patel	Both HR users added	Added successfully	Pass
17	HR Selection Panel	Select HR	No HR selected		Validation Error	Validation Error	Pass
18	Reviewer Selection Panel	Search Employees	Search employee by name	CEO	Employee displayed in list	Employee displayed	Pass
19	Reviewer Selection Panel	Search Employees	Search employee by email	vp@gmail.com	Employee displayed	Employee displayed	Pass
20	Reviewer Selection Panel	Select Reviewer	Add reviewer using + button	VP	Reviewer added to selected list	Reviewer added	Pass

Test Case ID	Form/Screen	Field Name	Test Scenario	Input Data	Expected Result	Actual Result	Status (Pass / Fail)
21	Reviewer Selection Panel	Select Reviewer	Add multiple reviewers	VP, System Admin	Both reviewers added	Added successfully	Pass
22	Reviewer Selection Panel	Select Reviewer	No reviewer selected		Validation Error	Validation Error	Pass
23	Job Details Form	Submit Button	Submit with valid inputs	Valid job details	Job created successfully	Job created successfully	Pass
24	Job Details Form	Submit Button	Submit with missing fields	Missing title	Validation error displayed	Validation error displayed	Pass

3. Game Create Form:

Test Case ID	Form/Screen	Field Name	Test Scenario	Input Data	Expected Result	Actual Result	Status (Pass / Fail)
1	Create New Game	Name	Valid Game Name	Chess	Accepted	Accepted	Pass
2	Create New Game	Name	Empty Game Name		Validation Error	Validation Error	Pass
3	Create New Game	Name	Game name too long	300 characters	Validation Error	Validation Error	Pass
4	Create New Game	Max Player	Valid Max Player	4	Accepted	Accepted	Pass
5	Create New Game	Max Player	Empty Max Player		Validation Error	Validation Error	Pass
6	Create New Game	Max Player	Negative value	-2	Validation Error	Validation Error	Pass
7	Create New Game	Max Player	Non-numeric value	abc	Validation Error	Validation Error	Pass

Test Case ID	Form/Screen	Field Name	Test Scenario	Input Data	Expected Result	Actual Result	Status (Pass / Fail)
8	Create New Game	Min Player	Valid Min Player	2	Accepted	Accepted	Pass
9	Create New Game	Min Player	Empty Min Player		Validation Error	Validation Error	Pass
10	Create New Game	Min Player	Min Player greater than Max Player	Min=5 Max=4	Validation Error	Validation Error	Pass
11	Create New Game	Duration (Minutes)	Valid Duration	30	Accepted	Accepted	Pass
12	Create New Game	Duration (Minutes)	Empty Duration		Validation Error	Validation Error	Pass
13	Create New Game	Duration (Minutes)	Negative Duration	-10	Validation Error	Validation Error	Pass
14	Create New Game	Slot Assigned (Min)	Valid Slot Assigned	60	Accepted	Accepted	Pass
15	Create New Game	Slot Assigned (Min)	Empty Slot Assigned		Validation Error	Validation Error	Pass
16	Create New Game	Slot Assigned (Min)	Non-numeric value	abc	Validation Error	Validation Error	Pass
17	Create New Game	Slot Create For Next (Days)	Valid days value	7	Accepted	Accepted	Pass
18	Create New Game	Slot Create For Next (Days)	Empty days value		Validation Error	Validation Error	Pass
19	Create New Game	Slot Create For Next (Days)	Negative days value	-5	Validation Error	Validation Error	Pass

Test Case ID	Form/Screen	Field Name	Test Scenario	Input Data	Expected Result	Actual Result	Status (Pass / Fail)
20	Create New Game	Create Slot Button	Submit valid game configuration	Valid form data	Game created and slots generated successfully	Game created successfully	Pass

4. Post Create Form:

Test Case ID	Form/Screen	Field Name	Test Scenario	Input Data	Expected Result	Actual Result	Status (Pass / Fail)
1	Create New Post	Title	Valid Title	Team Achievement	Accepted	Accepted	Pass
2	Create New Post	Title	Empty Title		Validation Error	Validation Error	Pass
3	Create New Post	Title	Very long title	300 characters	Validation Error	Validation Error	Pass
4	Create New Post	Description	Valid Description	Our team successfully completed the project	Accepted	Accepted	Pass
5	Create New Post	Description	Empty Description		Validation Error	Validation Error	Pass
6	Create New Post	Description	Very long description	1000+ characters	Accepted	Accepted	Pass
7	Create New Post	Image	Valid image upload	post.jpg	Accepted	Accepted	Pass
8	Create New Post	Image	Upload PNG image	post.png	Accepted	Accepted	Pass

Test Case ID	Form/Screen	Field Name	Test Scenario	Input Data	Expected Result	Actual Result	Status (Pass / Fail)
9	Create New Post	Image	Upload invalid file type	post.pdf	Validation Error	Validation Error	Pass
10	Create New Post	Image	No image uploaded		Validation Error	Validation Error	Pass
11	Create New Post	Tags	Add valid tag	celebration	Tag added successfully	Tag added	Pass
12	Create New Post	Tags	Add multiple tags	celebration, success	Tags added successfully	Tags added	Pass
13	Create New Post	Tags	Empty tag input		Tag not added	Tag not added	Pass
14	Create New Post	Public Checkbox	Checkbox checked	True	Post will be public	Post marked public	Pass
15	Create New Post	Public Checkbox	Checkbox unchecked	False	Post will be private	Post marked private	Pass
16	Create New Post	Create Post Button	Submit valid post	Valid form data	Post created successfully	Post created successfully	Pass
17	Create New Post	Create Post Button	Submit with missing fields	Missing title	Validation error displayed	Validation error displayed	Pass
18	Create New Post	Cancel Button	Click cancel button	—	Form closed without saving	Form closed	Pass

5. User Create Form:

Test Case ID	Form/Screen	Field Name	Test Scenario	Input Data	Expected Result	Actual Result	Status (Pass / Fail)
1	Add New User	Full Name	Valid full name	Kartik Patel	Accepted	Accepted	Pass
2	Add New User	Full Name	Empty full name		Validation Error	Validation Error	Pass
3	Add New User	Full Name	Name with numbers	Kartik123	Validation Error	Validation Error	Pass
4	Add New User	Email	Valid email	user@gmail.com	Accepted	Accepted	Pass
5	Add New User	Email	Empty email		Validation Error	Validation Error	Pass
6	Add New User	Email	Invalid email format	usergmail.com	Validation Error	Validation Error	Pass
7	Add New User	Email	Existing email	hr@gmail.com	Server Side Error	Server Side Error	Pass
8	Add New User	Password	Valid password	User@123	Accepted	Accepted	Pass
9	Add New User	Password	Empty password		Validation Error	Validation Error	Pass
10	Add New User	Password	Weak password	12345	Validation Error	Validation Error	Pass
11	Add New User	Designation	Valid designation	Software Engineer	Accepted	Accepted	Pass
12	Add New User	Designation	Empty designation		Validation Error	Validation Error	Pass
13	Add New User	Role	Select valid role	Employee	Accepted	Accepted	Pass
14	Add New User	Role	No role selected		Validation Error	Validation Error	Pass

Test Case ID	Form/Screen	Field Name	Test Scenario	Input Data	Expected Result	Actual Result	Status (Pass / Fail)
15	Add New User	Department	Select valid department	IT	Accepted	Accepted	Pass
16	Add New User	Department	No department selected		Validation Error	Validation Error	Pass
17	Add New User	Date of Birth	Valid date	10-05-2000	Accepted	Accepted	Pass
18	Add New User	Date of Birth	Empty DOB		Validation Error	Validation Error	Pass
19	Add New User	Date of Join	Valid joining date	01-07-2024	Accepted	Accepted	Pass
20	Add New User	Date of Join	Joining date before DOB	DOB: 10-05-2000 Join: 01-01-1995	Validation Error	Validation Error	Pass
21	Add New User	Report To	Select manager	VP	Accepted	Accepted	Pass
22	Add New User	Profile Image	Upload valid image	profile.jpg	Accepted	Accepted	Pass
23	Add New User	Profile Image	Upload invalid file	profile.pdf	Validation Error	Validation Error	Pass
24	Add New User	Profile Image	No image uploaded		Validation Error	Validation Error	Pass
25	Add New User	Create User Button	Submit valid form	Valid form data	User created successfully	User created successfully	Pass
26	Add New User	Cancel Button	Click cancel button	—	Form closes without saving	Form closed	Pass

6. Expense Create Form:

Test Case ID	Form/Screen	Field Name	Test Scenario	Input Data	Expected Result	Actual Result	Status (Pass / Fail)
1	Add New Expense	Details	Valid expense details	Taxi fare to airport	Accepted	Accepted	Pass
2	Add New Expense	Details	Empty expense details		Validation Error	Validation Error	Pass
3	Add New Expense	Details	Very long description	500+ characters	Accepted	Accepted	Pass
4	Add New Expense	Amount	Valid amount	1500	Accepted	Accepted	Pass
5	Add New Expense	Amount	Empty amount		Validation Error	Validation Error	Pass
6	Add New Expense	Amount	Negative amount	-100	Validation Error	Validation Error	Pass
7	Add New Expense	Amount	Non-numeric amount	abc	Validation Error	Validation Error	Pass
8	Add New Expense	Expense Date	Valid expense date	10-06-2026	Accepted	Accepted	Pass
9	Add New Expense	Expense Date	Empty expense date		Validation Error	Validation Error	Pass
10	Add New Expense	Expense Date	Date outside travel period	01-01-2030	Validation Error	Validation Error	Pass
11	Add New Expense	Category	Valid category selected	Travel	Accepted	Accepted	Pass
12	Add New Expense	Category	No category selected		Validation Error	Validation Error	Pass
13	Add New Expense	Proof	Upload valid PDF receipt	receipt.pdf	Accepted	Accepted	Pass
14	Add New Expense	Proof	Upload valid image receipt	receipt.jpg	Accepted	Accepted	Pass

Test Case ID	Form/Screen	Field Name	Test Scenario	Input Data	Expected Result	Actual Result	Status (Pass / Fail)
15	Add New Expense	Proof	Upload unsupported file type	receipt.exe	Validation Error	Validation Error	Pass
16	Add New Expense	Proof	No file uploaded		Validation Error	Validation Error	Pass
17	Add New Expense	Submit Button	Submit with valid form data	Valid expense data	Expense submitted successfully	Expense submitted successfully	Pass
18	Add New Expense	Submit Button	Submit with missing required fields	Missing amount	Validation error displayed	Validation error displayed	Pass

7. Travel Document Add Form:

Test Case ID	Form/Screen	Field Name	Test Scenario	Input Data	Expected Result	Actual Result	Status (Pass / Fail)
1	Add Travel Document	Document Name	Valid document name	Flight Ticket	Accepted	Accepted	Pass
2	Add Travel Document	Document Name	Empty document name		Validation Error	Validation Error	Pass
3	Add Travel Document	Document Name	Very long document name	300 characters	Validation Error	Validation Error	Pass
4	Add Travel Document	Document Type	Valid document type	Ticket	Accepted	Accepted	Pass
5	Add Travel Document	Document Type	Empty document type		Validation Error	Validation Error	Pass
6	Add Travel Document	Document Type	Invalid characters in	@@@	Validation Error	Validation Error	Pass

Test Case ID	Form/Screen	Field Name	Test Scenario	Input Data	Expected Result	Actual Result	Status (Pass / Fail)
			document type				
7	Add Travel Document	Document File	Upload valid PDF file	ticket.pdf	Accepted	Accepted	Pass
8	Add Travel Document	Document File	Upload image file	ticket.jpg	Accepted	Accepted	Pass
9	Add Travel Document	Document File	Upload unsupported file type	ticket.exe	Validation Error	Validation Error	Pass
10	Add Travel Document	Document File	No file uploaded		Validation Error	Validation Error	Pass
11	Add Travel Document	Upload Document Button	Submit valid document details	Valid document data	Document uploaded successfully	Document uploaded successfully	Pass
12	Add Travel Document	Upload Document Button	Submit with missing required fields	Missing document name	Validation error displayed	Validation error displayed	Pass
13	Add Travel Document	Cancel Button	Click cancel button	—	Modal closed without saving	Modal closed	Pass

9 . FUTURE ENHANCEMENT

- Mobile Application Support
- Multi-Level Approval Workflow
- AI-Based Resume Screening
- Real-Time Notifications System

10 . BIBLIOGRAPHY